OSCILLATION & WAVES

1. INTRODUCTION

(1) A motion which repeats itself over and over again after a regular interval of time is called a periodic motion.

(2) Oscillatory or vibratory motion is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point in a definite interval of time.

(3) Simple harmonic motion is a specific type of oscillatory motion, in which

(a) partical moves in one dimension,

(b) particle moves to and fro about a fixed mean position (where $F_{net} = 0$),

(c) net force on the particle is always directed towards means position, and

(d) magnitude of net force is always proportional to the displacement of particle from the mean position at that instant.

So, $F_{net} = -kx$

where, k is known as force constant

$\Rightarrow$ $ma = -kx$

$\Rightarrow$ $a = \frac{-k}{m}x$ or $a = -\omega^2 x$

where, ω is known as angular frequency.

$\Rightarrow$ $\frac{d^2x}{dt^2} = -\omega^2 x$

This equation is called as the differential equation of S.H.M.

The general expression for x(t) satisfying the above equation is :

$x(t) = A \sin(\omega t + \phi)$

1.1 Some Important terms

1. Amplitude

The amplitude of particle executing S.H.M. is its maximum displacement on either side of the mean position.

A is the amplitude of the particle.

2. Time Period

Time period of a particle executing S.H.M. is the time taken to complete one cycle and is denoted by T.

Time period $(T) = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$ as $\omega = \sqrt{\frac{m}{k}}$

3. Frequency

The frequency of a particle executing S.H.M. is equal to the number of oscillations completed in one second.

$v = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

4. Phase

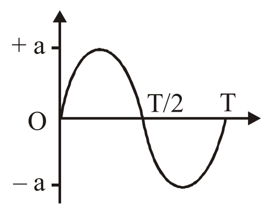
The phase of particle executing S.H.M. at any instant is its state as regard to its position and direction of motion at that instant. it is measured as argument (angle) of sine in the equation of S.H.M.

Phase = $(\omega t + \phi)$

At $t = 0$, phase = ϕ; the constant ϕ is called initial phase of the particle or phase constant.

1.2 Important Relations

1. Position

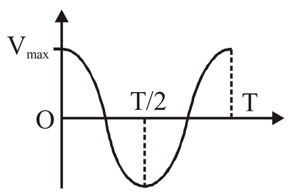


If mean position is at origin the position (X coordinate) depends on time in general as :

$x(t) = \sin(\omega t + \phi)$

- At mean position, $x = 0$
- At extremes, $x = +A, -A$

2. Velocity



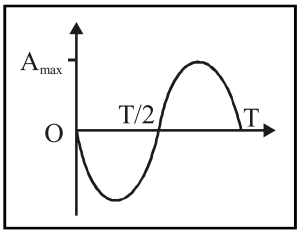
- At any time instant t, $v(t) = A\omega\cos(\omega t + \phi)$

- At any position x, $v(x) = \pm\, \omega\sqrt{A^2 - x^2}$
- Velocity is minimum at extremes because the particles is at rest.

  i.e., $v = 0$ at extreme position.
- Velocity has maximum magnitude at mean position.

  $|v|_{max} = \omega A$ at mean position.

3. Acceleration



- At any instant t, $a(t) = -\omega^2 A \sin(\omega t + \phi)$
- At any position x, $a(x) = -\omega^2 x$
- Acceleration is always directed towards mean position.
- The magnitude of acceleration is minimum at mean position and maximum at extremes.

  $|a|_{min} = 0$ at mean position.

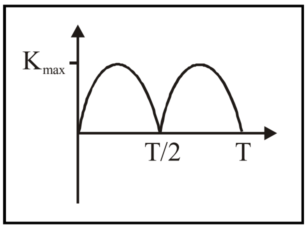
  $|a|_{max} = \omega^2 A$ at extremes.

4. Energy

<u>Kinetic energy</u>

- $K = \frac{1}{2}mv^2 \Rightarrow K = \frac{1}{2}m\omega^2\left(A^2 - x^2\right)$

  $= \frac{1}{2}m\omega^2 A^2 \cos^2(\omega t + \phi)$

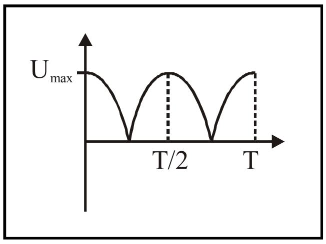


- K is maximum at mean position and minimum at extremes.
- $K_{max} = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2$ at mean position
- $K_{min} = 0$ at extemes.

<u>Potential Energy</u>

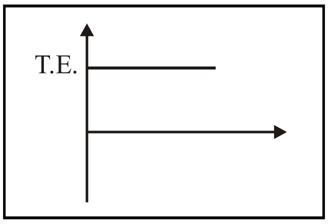
If potential energy is taken as zero at mean position, then at any position x,

$$U(x) = \frac{1}{2}kx^2 = \frac{1}{2}mA^2\omega^2 \sin^2(\omega t + \phi)$$



- U is maximum at extremes $U_{max} = \frac{1}{2}kA^2$
- U is minimum at mean position

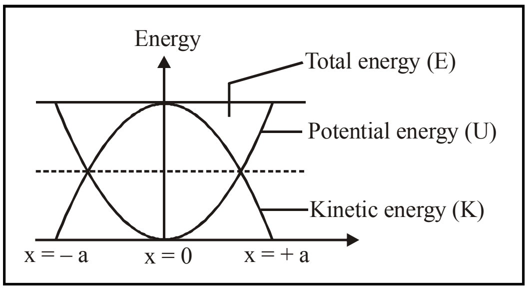
<u>Total Energy</u>



$$T.E. = \frac{1}{2}kA^2 = \frac{1}{2}mA^2\omega^2$$

and is constant at all time instant and at all positions.

Energy position graph



2. TIME PERIOD OF S.H.M.

To find whether a motion is S.H.M. or not and to find its time period, follow these steps :

(a) Locate the mean (equilibrium) position mathematically by balancing all the forces on it.

(b) Displace the particle by a displacement 'x' from the mean position in the probable direction of oscillation.

(c) Find the net force on it and check if it is towards mean position.

(d) Try to express net force as a proportional function of its displacement 'x'.

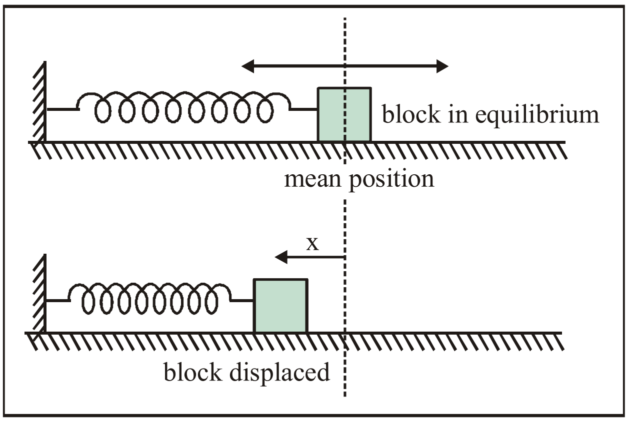
✹ If step (c) and step (d) are proved then it is a simple harmonic motion.

(e) Find k from expression of net force (F = – kx) and find time period using $T = 2\pi\sqrt{\frac{m}{k}}$.

2.1 Oscillations of a Block Connected to a Spring

(a) Horizontal spring :

Let a block of mass m be placed on a smooth horizontal surface and rigidly connected to spring of force constant K whose other end is permanently fixed.

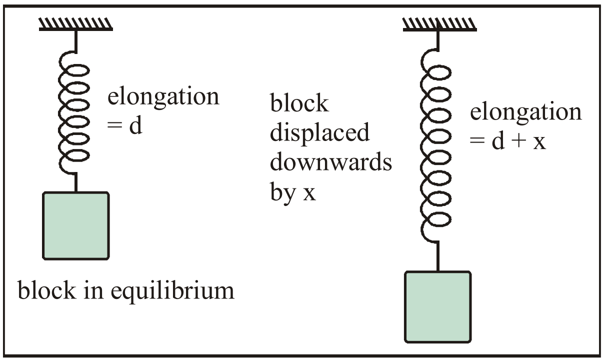


✹ Mean position : when spring is at its natural length.

✹ Time period : $T = 2\pi\sqrt{\frac{m}{k}}$

(b) Vertical Spring :

If the spring is suspended vertically from a fixed point and carries the block at its other end as shown, the block will oscillate along the vertical line.



✹ Mean position : spring in elongated by $d = \frac{mg}{k}$

✹ Time period : $T = 2\pi\sqrt{\frac{m}{k}}$

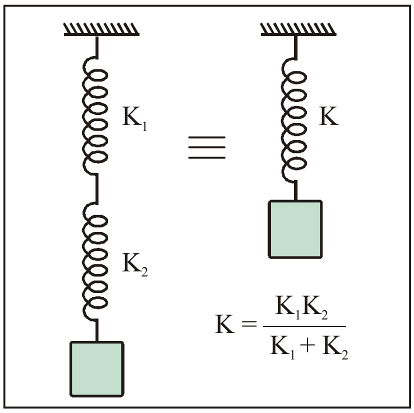
(c) Combination of springs :

1. Springs in series

When two springs of force constant K_1 and K_2 are connected in series as shown, they are equivalent to a single spring of force constant K which is given by

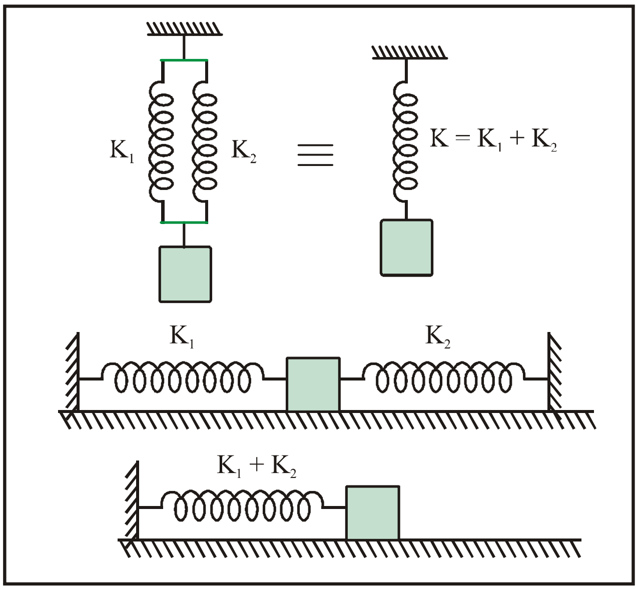
$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2}$$

$$K = \frac{K_1K_2}{K_1 + K_2}$$



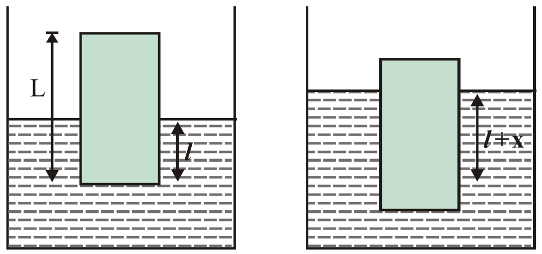
2. Springs in parallel

For a parallel combination as shown, the effective spring constant is $K = K_1 + K_2$



2.2 Oscillation of a Cylinder Floating in a liquid

Let a cylinder of mass m and density d be floating on the surface of a liquid of density ρ. The total length of cylinder is L.

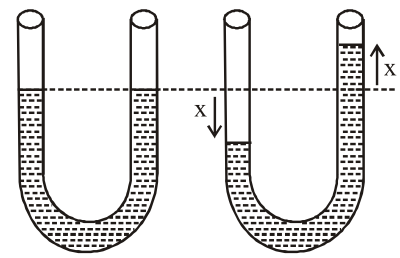


cylinder in equilibrium   cylinder displaced downwards by a distance x

- Mean position : cylinder is immersed upto $\ell = \frac{Ld}{\rho}$
- Time period : $T = 2\pi\sqrt{\frac{Ld}{\rho g}} = 2\pi\sqrt{\frac{\ell}{g}}$

2.3 Liquid Oscillating in a U–Tube

Consider a liquid column of mass m and density ρ in a U-tube of area of cross section A.

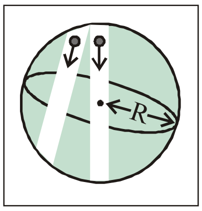


In equilibrium   liquid displaced by x

- Mean position : when height of liquid is same in both limbs.
- Time period : $T = 2\pi\sqrt{\frac{m}{2A\rho g}} = 2\pi\sqrt{\frac{L}{2g}}$

where, L is length of liquid column.

2.4 Body Oscillation in tunnel along any chord of earth



- Mean position : At the centre of the chord
- Time period : $T = 2\pi\sqrt{\frac{R}{g}} = 84.6$ minutes

where, R is radius of earth.

2.5 Angular Oscillations

Instead of straight line motion, if a particle or centre of mass of a body is oscillating on a small arc of circular path then it is called angular S.H.M.

For angular S.H.M., $\tau = -k\theta$

$\Rightarrow$ $I\alpha = -k\theta$

$\Rightarrow$ Time period, $T = 2\pi\sqrt{\frac{I}{k}}$

2.5.1 Simple Pendulum

- Time period : $T = 2\pi\sqrt{\frac{\ell}{g}}$
- Time period of a pendulum in a lift :

$T = 2\pi\sqrt{\frac{\ell}{g+a}}$ (if acceleration of lift is upwards)

$T = 2\pi\sqrt{\frac{\ell}{g-a}}$ (if acceleration of lift is downwards)

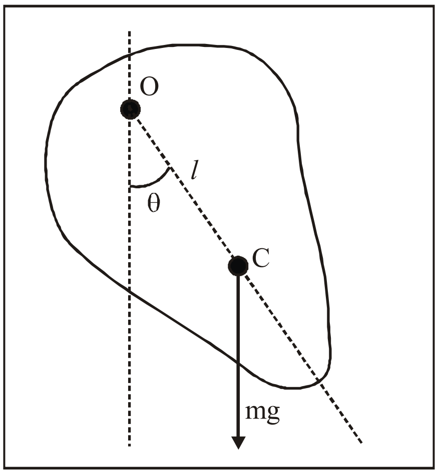
- Second's pendulum

Time period of second's pendulum is 2s.

Length of second's pendulum on earth surface $\simeq$ 1m.

2.5.2 Physical Pendulum

Time period : $T = 2\pi\sqrt{\frac{I}{mg\ell}}$



where, I is moment of inertia of object about point of suspension, and

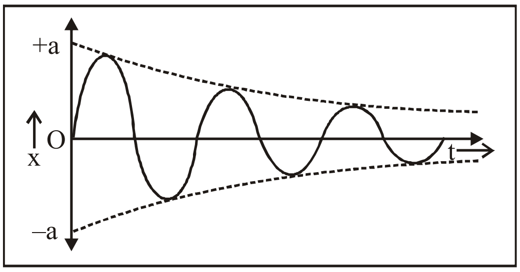
l is distance of centre of mass of object from point of suspension.

3. DAMPED AND FORCED OSCILLATIONS

1. Damped Oscillation :

(i) The oscillation of a body whose amplitude goes on decreasing with time is defined as damped oscillation.

(ii) In this oscillation the amplitude of oscillation decreases exponentially due to damping forces like frictional force, viscous force etc.



(iii) Due to decrease in amplitude the energy of the oscillator also goes on decreasing exponentially.

2. Forced Oscillation :

(i) The oscillation in which a body oscillates under the influence of an external periodic force are known as forced oscillation.

(ii) Resonance : When the frequency of external force is equal to the natural frequency of the oscillator, then this state is known as the state of resonance. And this frequency is known as resonant frequency.

4. WAVES

(a) Speed of longitudinal wave

- Speed of longitudinal wave in a medium is given by

$$v = \sqrt{\frac{E}{\rho}}$$

where, E is the modulus of elasticity,

ρ is the density of the medium.

- Speed of longitudinal wave in a solid in the form of rod is given by

$$v = \sqrt{\frac{Y}{\rho}}$$

where, Y is the Young's modulus of the solid,

ρ is the density of the solid.

- Speed of longitudinal wave in fluid is given by

$$v = \sqrt{\frac{B}{\rho}}$$

where, B is the bulk modulus,

ρ is the density of the fluid.

(b) Newton's formula

- Newton assumed that propagation of sound wave in gas is an isothermal process. Therefore, according to Newton, speed of sound in gas is given by $v = \sqrt{\frac{P}{\rho}}$

where P is the pressure of the gas and ρ is the density of the gas.

- According to the Newton's formula, the speed of sound in air at S.T.P. is 280 m/s. But the experimental value of the speed of sound in air is 332 m s^{-1}. Newton could not explain this large difference. Newton's formula was corrected by Laplace.

(c) Laplace's correction

- Laplace assumed that propagation of sound wave in gas in an adiabatic process. Therefore, according to Laplace, speed of sound in a gas is given by

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

- According to Laplace's correction the speed of sound in air at S.T.P. is 331.3 m/s. This value agrees farily well with the experimental values of the velocity of sound in air at S.T.P.

5. WAVES TRAVELLING IN OPPOSITE DIRECTIONS

- When two waves of same amplitude and frequency travelling in opposite directions

$$y_1 = A \sin (kx - \omega t)$$

$$y_2 = A \sin (kx + \omega t)$$

interfere, then a standing wave is produced which is given by,

$$y = y_1 + y_2$$

$$\Rightarrow \; y = 2A \sin kx \cos \omega t$$

- Hence the particle at location x is oscillating in S.H.M. with angular frequency ω and amplitude 2A sin kx. As the amplitude depends on location (x), particles are oscillating with different amplitude.

- **Nodes :** Amplitude = 0

  $2A \sin kx = 0$

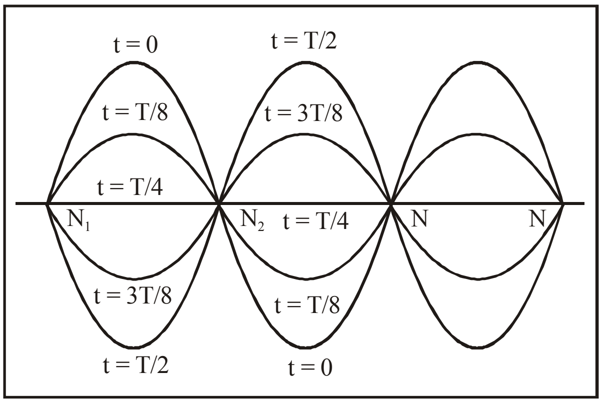
  $x = 0, \pi/k, 2\pi/k.......$

  $x = 0, \lambda/2, \lambda, 3\lambda/2, 2\lambda........$
- **Antinodes :** Amplitude is maximum.

  $\sin kx = \pm 1$

  $x = \pi/2k, 3\pi/2k$

  $x = \lambda/4, 3\lambda/4, 5\lambda/4$
- Nodes are completely at rest. Antinodes are oscillating with maximum amplitude (2A). The points between a node and antinode have amplitude between 0 and 2 A.
- Separation between two consecutive (or antinodes) $= \lambda/2$.
- Separation between a node and the next antinode$=\lambda/4$.
- Nodes and antinodes are alternately placed.



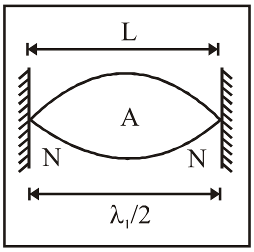
- It is clear from the figure that since nodes are, at rest they don't transfer energy. In a stationary wave, energy is not transferred from one point to the other.

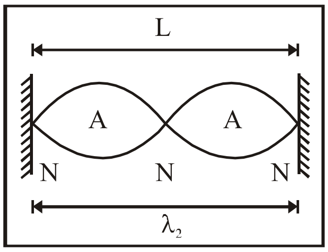
5.1 Vibrations in a stretched string

1. Fixed at both ends.

- Transverse standing waves with nodes at both ends of the string are formed.
- So, length of string, $\ell = \dfrac{n\lambda}{2}$ if there are (n + 1) nodes and n antinodes.
- Frequency of oscillations is

$$\Rightarrow \nu = \frac{v}{\lambda} = \frac{nv}{2\ell}$$





- Fundamental frequency (x = 1)

$$\nu_0 = \frac{v}{2L}$$

It is also called first harmonic.

- Second harmonic or first overtone

$$\nu = \frac{2v}{2L}$$

- The nth multiple of fundamental frequency is known as nth harmonic or (n – 1)th overtone.

2. Fixed at one end

- Transverse standing waves with node at fixed end and antinode at open end are formed.
- So, length of string $\ell = (2n-1)\dfrac{\lambda}{4}$ if there are n nodes and n antinodes.
- Frequency of oscillations

$$\Rightarrow \nu = \frac{v}{\lambda} = \frac{(2n-1)v}{4\ell}$$

- Fundamental frequency, (n = 1)

$$\nu_0 = \frac{v}{4L}$$

It is also called first harmonic.

- First overtone or third harmonic.

$$\nu = \frac{3v}{4\ell} = 3\nu_0$$

- Only odd harmonics are possible in this case.

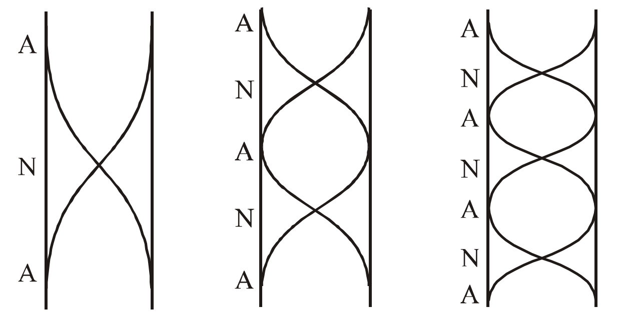
5.2 Vibrations in an organ pipe

1. Open Organ pipe (both ends open)

- The open ends of the tube becomes antinodes because the particles at the open end can oscillate freely.
- If there are (n + 1) antinodes in all,

length of tube, $\ell = \frac{n\lambda}{2}$

- So, Frequency of oscillations is $\nu = \frac{nv}{2\ell}$



fundamental mode first harmonic | Second harmonic (first overtone) | Third harmonic (second overtone)

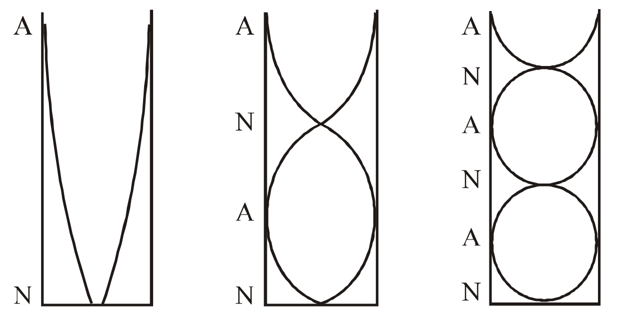
2. Closed organ pipe (One end closed)

- The open end becomes antinode and closed end become a node.
- If there are n nodes and n antinodes,

  L = (2n – 1) λ/4

- So frequency of oscillations is

$$\nu = \frac{v}{\lambda} = \frac{(2n-1)v}{4L}$$



fundamental mode first harmonic | third harmonic (first overtone) | fifth harmonic (second overtone)

- There are only odd harmonics in a tube closed at one end.

5.3 Waves having different frequencies

Beats are formed by the superposition of two waves of slightly different frequencies moving in the same direction. The resultant effect heard in this case at any fixed position will consist of alternate loud and weak sounds.

Let us consider net effect of two waves of frequencies ν_1 and ν_2 and amplitude A at x = 0.

$$y_1 = A\sin 2\pi\nu_1 t$$

$$y_2 = A\sin 2\pi\nu_2 t$$

$\Rightarrow y = y_1 + y_2$

- $\Rightarrow y = A\left(\sin 2\pi\nu_1 t + \sin 2\pi\nu_2 t\right)$

$$y = \left[2A\cos\pi(\nu_1 - \nu_2)t\right]\sin\pi(\nu_1 + \nu_2)t$$

Thus the resultant wave can be represented as a travelling wave whose frequency is $\left(\frac{\nu_1 + \nu_2}{2}\right)$ and amplitude is $2A\cos\pi(\nu_1 - \nu_2)t$.

As the amplitude term contains t, the amplitude varies periodically with time.

For Loud Sounds : Net amplitude = ± 2A

$\Rightarrow \cos\pi(\nu_1 - \nu_2)t = \pm 1$

$\Rightarrow \pi(\nu_1 - \nu_2)t = 0, \pi, 2\pi, 3\pi$ ....

$\Rightarrow t = 0, \frac{1}{\nu_1 - \nu_2}, \frac{2}{\nu_1 - \nu_2}$ ....

Hence the interval between two loud sounds is given as :

$$= \frac{1}{\nu_1 - \nu_2}$$

$\Rightarrow$ the number of loud sounds per second = $\nu_1 - \nu_2$

$\Rightarrow$ beat per second = $\nu_1 - \nu_2$

Note that $\nu_1 - \nu_2$ must be small (0 – 16 Hz) so that sound variations can be distinguished.

Note...

- Filling a tuning form increases its frequency of vibration.
- Loading a tuning for k decreases its frequency of vibration.

6. DOPPLER EFFECT

According to Doppler's effect, whenever there is a relative motion between a source of sound and listener, the apparent frequency of sound heard by the listener is different from the actual frequency of sound emitted by the source.

Apparent frequency,

$$\boxed{\nu' = \frac{v - v_L}{v - v_s} \times \nu}$$

Sing Convention. All velocities along the direction S to L are taken as positve and all velocities along the direction L to S are taken as negative.

When the motion is along some other direction the component of velocity of source and listener along the line joining the source and listener is considered.

Special Cases :

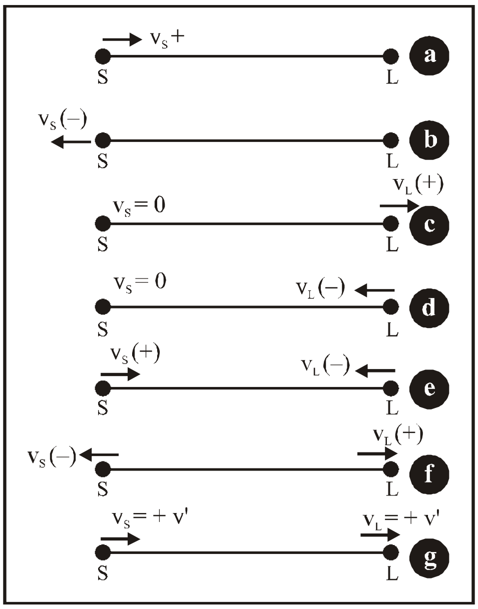
(a) If the source is moving towards the listener but the listener is at rest, then v_s is positive and $v_L = 0$ (figure a). Therefore,

$$\boxed{\nu' = \frac{v}{v - v_s} \times \nu} \text{ i.e. } \nu' > \nu$$

(b) If the source is moving away from the listener, but the listener is at rest, then v_s is negative and $v_L = 0$ (figure b). Therefore,

$$\boxed{\nu' = \frac{v}{v-(-v_s)}\nu = \frac{v}{v+v_s}\nu} \text{ i.e. } \nu' < \nu$$

(c) If the source is at rest and listener is moving away from the source, the $v_s = 0$ and v_L is positive (figure c). Therefore,



$$\boxed{\nu' = \frac{(v - v_L)}{v}\nu} \text{ i.e. } \nu' < \nu$$

(d) If the source is at rest and listener is moving towards the source, then $v_s = 0$ and v_L is negative (figure d). Therefore,

$$\boxed{\nu' = \frac{v-(-v_L)}{v}\nu = \frac{v+v_L}{v}\nu} \text{ i.e. } \nu' > \nu$$

(e) If the source and listener are approaching each other, then v_s is positive and v_L is negative (figure e). Therefore,

$$\boxed{\nu' = \frac{v-(-v_L)}{v-v_s}\nu = \left(\frac{v+v_L}{v-v_s}\right)\nu} \text{ i.e. } \nu' > \nu$$

(f) If the source and listener are moving away from each other, then v_s is negative and v_L is positive, (figure f). Therefore,

$$\boxed{\nu' = \frac{v-v_L}{v-(-v_s)}\nu = \frac{v-v_L}{v+v_s}\nu} \text{ i.e. } \nu' < \nu$$

(g) If the source and listener are both in motion in the same direction and with same velocity, then $v_s = v_L = v'$ (say) (figure g). Therefore,

$$\boxed{\nu' = \frac{(v-v')}{(v-v')}\nu} \text{ i.e. } \nu' = \nu$$

It means, there is no change in the frequency of sound heard by the listerner.

Apparent wavelength heard by the observer is

$$\lambda' = \frac{v - v_s}{\nu}$$

Note...

If case the medium is also moving, the speed of sound with respect to ground is considered. i.e. $\vec{v} + \vec{v}_m$

7. CHARACTERISTICS OF SOUND

- **Loudness** of sound is also called level of intensity of sound.

  In decibel the loudness of a sound of intensity I is given by $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$. ($I_0 = 10^{-12}$ w/m^2)

- **Pitch** : It is pitch depends on frequency, higher the frequency higher will be the pitch and shriller will be the sound.

Physics Notes Class 11 CHAPTER 13 KINETIC THEORY

Assumptions of Kinetic Theory of Gases

1. Every gas consists of extremely small particles known as molecules. The molecules of a given gas are all identical but are different from those of another gas.
2. The molecules of a gas are identical spherical, rigid and perfectly elastic point masses.
3. Their molecular size is negligible in comparison to intermolecular distance (10^{-9} m).
4. The speed of gas molecules lies between zero and infinity (very high speed).
5. The distance covered by the molecules between two successive collisions is known as free path and mean of all free path is known as mean free path.
6. The number of collision per unit volume in a gas remains constant.
7. No attractive or repulsive force acts between gas molecules.
8. Gravitational to extremely attraction among the molecules is ineffective due small masses and very high speed of molecules.

Gas laws

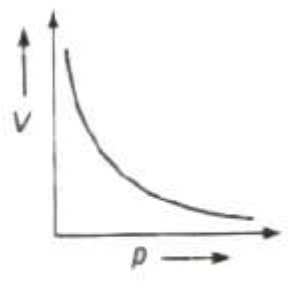
Assuming permanent gases to be ideal, through experiments, it was established that gases irrespective of their nature obey the following laws.

Boyle's Law

At constant temperature the volume (V) of given mass of a gas is inversely proportional to its pressure (p), i.e.,

$V \propto 1/p \Rightarrow pV = \text{constant}$

For a given geas, $p_1V_1 = p_2V_2$



Charles' Law

At constant pressure the volume (V) of a given mass of gas is directly proportional to its absolute temperature (T), i.e.,

$V \propto T \Rightarrow V / T$ = constant

For a given gas, $V_1/T_1 = V_2/T_2$

At constant pressure the volume (V) of a given mass of a gas increases or decreases by 1/273.15 of its volume at 0°C for each 1°C rise or fall in temperature.

Volume of the gas at t°Ce

$V_t = V_0 (1 + t/273.15)$

where V_0 is the volume of gas at 0°C.

Gay Lussacs' or Regnault's Law

At constant volume the pressure p of a given mass of gas is directly proportional to its absolute temperature T, i.e. ,

$p \propto T \Rightarrow V/T$ = constant

For a given gas,
$p_1/T_1 = p_2/T_2$

At constant volume (V) the pressure p of a given mass of a gas increases or decreases by 1/273.15 of its pressure at 0°C for each l°C rise or fall in temperature.

Volume of the gas at t°C, $p_t = p_0 (1 + t/273.15)$

where P_0 is the pressure of gas at 0°C.

Avogadro's Law

Avogadro stated that equal volume of all the gases under similar conditions of temperature and pressure contain equal number molecules. This statement is called Avogadro's hypothesis. According Avogadro's law

(i) Avogadro's number The number of molecules present in 1g mole of a gas is defined as Avogadro's number.

$N_A = 6.023 \text{ X } 10^{23}$ per gram mole

(ii) At STP or NTP (T = 273 K and p = 1 atm 22.4 L of each gas has 6.023×10^{23} molecules.

(iii) One mole of any gas at STP occupies 22.4 L of volume.

Standard or Perfect Gas Equation

Gases which obey all gas laws in all conditions of pressure and temperature are called perfect gases.

Equation of perfect gas pV=nRT

where p = pressure, V = volume, T = absolute temperature, R = universal gas constant and n = number of moles of a gas.

Universal gas constant R = 8.31 J $mol^{-1}K^{-1}$.

Real Gases

Real gases deviate slightly from ideal gas laws because

- Real gas molecules attract one another.
- Real gas molecules occupy a finite volume.

Real or Van der Waal's Gas Equation

$(p + a/V^2) (V – b) = RT$

where a and b are called van der Waals' constants.

Pressure due to an ideal gas is given by
$p = (1/3).(mn/V). c^2 = 1/3 \rho c^2$

For one mole of an ideal gas
$P = (1/3).(M/V).c^2$

where, m = mass of one molecule, n = number of molecules, V = volume of gas, $c = (c_1^2 + c_2^2 + \ldots + c_n^2) / n$ allde root mean square (rrns) velocity of the gas molecules and M = molecular weight of the gas. If p is the pressure of the gas and E is the kinetic energy per unit volume is E, then

$p = (2/3).E$

Kinetic Energy of a Gas

(i) Average kinetic energy of translation per molecule of a gas is given by

$E = (3/2)$ kt

where k = Boltzmann's constant.

(ii) Average kinetic energy of translation per mole of a gas is given by

E = (3/2) Rt

where R = universal gas constant.

(iii) For a given gas kinetic energy

$E \propto T \Rightarrow E_1/E_2 = T_1/T_2$

(iv) Root mean square (rms) velocity of the gas molecules is given by

$$c = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3p}{\rho}}$$

(v) For a given gas $c \propto \sqrt{T}$

(vi) For different gases $c \propto 1/\sqrt{M}$

(vii) Boltzmann's constant k = R/N

where R is ideal gas constant and N = Avogadro number.

Value of Boltzmann's constant is 1.38×10^{-28} J/K.

(viii) The average speed of molecules of a gas is given by

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}} = \sqrt{\frac{8RT}{\pi M}}$$

(ix) The most probable speed of molecules of a gas is given by

$$v_{mp} = \sqrt{\frac{2kT}{m}} = \sqrt{\frac{2RT}{M}}$$

$\Rightarrow$ $$v_{rms} > \overline{v} > v_{mp}$$

Important Points

(i) With rise in temperature rms speed of gas molecules increases as

$$v_{rms} \propto \sqrt{T}$$

(ii) With the increase in molecular weight rms speed of gas molecule decrease as

$$v_{rms} \propto \frac{1}{\sqrt{M}}$$

(iii) Rms speed of gas molecules is of the order of km/s, *e.g.*, at NTP for hydrogen gas

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.31 \times 273}{2 \times 10^3}}$$

(iv) Rms speed of gas molecules does not depend on the pressure of gas (if temperature remains constant) because $p \propto \rho$ (Boyle's law). If pressure is increased n times, then density will also increase by n times but v_{rms} remains constant.

Degree of Freedom

The degree of freedom for a dynamic system is the number of directions in which it can move freely or the number of coordinates required to describe completely the position and configuration of the system.

It is denoted by for N.

Degree of freedom of a system is given by

f or N = 3A – R

where A = number of particles in the system and R = number of independent relations.

Degree of Freedom

1. For monoatomic gas = 3
2. For diatomic gas = 5
3. For non-linear triatomic gas = 6
4. For linear triatomic gas = 7

Specific heat of a gas

(a) At constant volume, $C_V = f/2\ R$

(b) At constant pressure, $c_p = (f/2 + 1)R$

(c) Ratio of specific heats of a gas at constant pressure and at constant volume is given by
$\gamma = 1 + 2/f$

Mean Free Path

The average distance travelled by a molecule between two successive collisions is called mean free path (γ).

Mean free path is given by

$\gamma = kT / \sqrt{2}\ \pi\ \sigma^2 p$

where σ = diameter of the molecule, p = pressure of the gas,
T = temperature and k = Botlzmann's constant.

Mean free path

$\lambda \propto T$ and $\lambda \propto 1/p$

Brownian Motion

The continuous random motion of the particles of microscopic size suspended in air or any liquid, is called Brownian of microscopic motion.

Brownian suspended motion in both is observed with many liquids and gases.

Brownian motion is due to the unequal bombardment of the suspended Particles by the molecules of the surrounding medium.

Physics Notes Class 11 CHAPTER 12 THERMODYNAMICS

The branch dealing with measurement of temperature is called thremometry and the devices used to measure temperature are called thermometers.

Heat

Heat is a form of energy called thermal energy which flows from a higher temperature body to a lower temperature body when they are placed in contact.

Heat or thermal energy of a body is the sum of kinetic energies of all its constituent particles, on account of translational, vibrational and rotational motion.

The SI unit of heat energy is joule (J).

The practical unit of heat energy is calorie.

1 cal = 4.18 J

1 calorie is the quantity of heat required to raise the temperature of 1 g of water by 1°C.

Mechanical energy or work (W) can be converted into heat (Q) by 1 W = JQ

where J = Joule's mechanical equivalent of heat.

J is a conversion factor (not a physical quantity) and its value is 4.186 J/cal.

Temperature

Temperature of a body is the degree of hotness or coldness of the body. A device which is used to measure the temperature, is called a thermometer.

Highest possible temperature achieved in laboratory is about 108 while lowest possible temperature attained is 10-8 K.

Branch of Physics dealing with production and measurement temperature close to 0 K is known as cryagenics, while that deaf with the measurement of very high temperature is called pyromet Temperature of the core of the sun is 107 K while that of its surface 6000 K.

NTP or STP implies 273.15 K (0°C = 32°F).

Different Scale of Temperature

1. **Celsius Scale** In this scale of temperature, the melting point ice is taken as 0°C and the boiling point of water as 100°C and space between these two points is divided into 100 equal parts
2. **Fahrenheit Scale** In this scale of temperature, the melt point of ice is taken as 32°F and the boiling point of water as 211 and the space between these two points is divided into 180 equal parts.
3. **Kelvin Scale** In this scale of temperature, the melting pouxl ice is taken as 273 K and the boiling point of water as 373 K the space between these two points is divided into 100 equal pss

Relation between Different Scales of Temperatures

$$\frac{C}{100} = \frac{F-32}{180} = \frac{K-273}{100} = \frac{R}{80}$$

Thermometric Property

The property of an object which changes with temperature, is call thermometric property. Different thermometric properties thermometers have been given below

(i) Pressure of a Gas at Constant Volume

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

and

$$p_t = p_0\left(1 + \frac{t}{273}\right)$$

$$t = \left(\frac{p_t - p_0}{p_{100} - p_0} \times 100\right) {}^\circ C$$

where p, p_{100}. and p_t, are pressure of a gas at constant volume 0°C, 100°C and t°C.

A constant volume gas thermometer can measure tempera from – 200°C to 500°C.

(ii) Electrical Resistance of Metals

$R_t = R_0(1 + \alpha t + \beta t^2)$

where α and β are constants for a metal.

As β is too small therefore we can take

$R_t = R_0(1 + \alpha t)$

where, α = temperature coefficient of resistance and
R_0 and R_t, are electrical resistances at 0°C and t°C.

$$\alpha = \frac{R_2 - R_1}{R_1 t_2 - R_2 t_1}$$

where R_1 and R_2 are electrical resistances at temperatures t_1 and t_2.

$$t = \frac{R_t - R_0}{R_{100} - R_0} \times 100°C$$

where R_{100} is the resistance at 100°C.

Platinum resistance thermometer can measure temperature from —200°C to 1200°C.

(iii) Length of Mercury Column in a Capillary Tube

$l_t = l_0(1 + \alpha t)$

where α = coefficient of linear expansion and l_0, l_t are lengths of mercury column at 0°C and t°C.

Thermo Electro Motive Force

When two junctions of a thermocouple are kept at different temperatures, then a thermo-emf is produced between the junctions, which changes with temperature difference between the junctions. Thermo-emf

$E = at + bt^2$

where a and b are constants for the pair of metals.

Unknown temperature of hot junction when cold junction is at 0°C.

$$t = \left(\frac{E_t}{E_{100}} \times 100\right)°C$$

Where E_{100} is the thermo-emf when hot junction is at 100°C.

A thermo-couple thermometer can measure temperature from —200°C to 1600°C.

Thermal Equilibrium

When there is no transfer of heat between two bodies in contact, the the bodies are called in thermal equilibrium.

Zeroth Law of Thermodynamics

If two bodies A and B are separately in thermal equilibrium with thirtli body C, then bodies A and B will be in thermal equilibrium with each other.

Triple Point of Water

The values of pressure and temperature at which water coexists inequilibrium in all three states of matter, i.e., ice, water and vapour
called triple point of water.

Triple point of water is 273 K temperature and 0.46 cm of mere pressure.

Specific Heat

The amount of heat required to raise the temperature of unit mass the substance through 1°C is called its specific heat.

It is denoted by c or s.

Its SI unit is joule/kilogram-°C'(J/kg-°C). Its dimensions is $[L^2T^{-2}\theta^{-1}]$.

The specific heat of water is 4200 J kg^{-1}°C^{-1} or 1 cal g^{-1} C^{-1}, which high compared with most other substances.

Gases have two types of specific heat

1. The specific heat capacity at constant volume (C_v).
2. The specific heat capacity at constant pressure (C_r).

Specific heat at constant pressure (C_p) is greater than specific heat constant volume (C_V), i.e., $C_p > C_V$.

For molar specific heats $C_p - C_V = R$
where R = gas constant and this relation is called Mayer's formula.

The ratio of two principal sepecific heats of a gas is represented by γ.

$$\gamma = \frac{C_p}{C_V}$$

The value of y depends on atomicity of the gas.

Amount of heat energy required to change the temperature of any substance is given by

$Q = mc\Delta t$

- where, m = mass of the substance,
- c = specific heat of the substance and
- Δt = change in temperature.

Thermal (Heat) Capacity

Heat capacity of any body is equal to the amount of heat energy required to increase its temperature through 1°C.

Heat capacity = me

where c = specific heat of the substance of the body and m = mass of the body.

Its SI unit is joule/kelvin (J/K).

Water Equivalent

It is the quantity of water whose thermal capacity is same as the heat capacity of the body. It is denoted by W.

W = ms = heat capacity of the body.

Latent Heat

The heat energy absorbed or released at constant temperature per unit mass for change of state is called latent heat.

Heat energy absorbed or released during change of state is given by

Q = mL

where m = mass of the substance and L = latent heat.

Its unit is cal/g or J/kg and its dimension is $[L^2T^{-2}]$.

For water at its normal boiling point or condensation temperature (100°C), the latent heat of vaporisation is

L = 540 cal/g
= 40.8 kJ/ mol
= 2260 kJ/kg

For water at its normal freezing temperature or melting point (0°C), the latent heat of fusion is

L = 80 cal/ g = 60 kJ/mol
= 336 kJ/kg

It is more painful to get burnt by steam rather than by boiling was 100°C gets converted to water at 100°C, then it gives out 536 heat. So, it is clear that steam at 100°C has more heat than wat 100°C (i.e., boiling of water).

After snow falls, the temperature of the atmosphere becomes very This is because the snow absorbs the heat from the atmosphere to down. So, in the mountains, when snow falls, one does not feel too but when ice melts, he feels too cold.

There is more shivering effect of ice cream on teeth as compare that of water (obtained from ice). This is because when ice cream down, it absorbs large amount of heat from teeth.

Melting

Conversion of solid into liquid state at constant temperature is melting.

Evaporation

Conversion of liquid into vapour at all temperatures (even below boiling point) is called evaporation.

Boiling

When a liquid is heated gradually, at a particular temperature saturated vapour pressure of the liquid becomes equal to atmospheric pressure, now bubbles of vapour rise to the surface d liquid. This process is called boiling of the liquid.

The temperature at which a liquid boils, is called boiling point The boiling point of water increases with increase in pre sure decreases with decrease in pressure.

Sublimation

The conversion of a solid into vapour state is called sublimation.

Hoar Frost

The conversion of vapours into solid state is called hoar fr..

Calorimetry

This is the branch of heat transfer that deals with the measorette heat. The heat is usually measured in calories or kilo calories.

Principle of Calorimetry

When a hot body is mixed with a cold body, then heat lost by ha is equal to the heat gained by cold body.

Heat lost = Heat gain

Thermal Expansion

Increase in size on heating is called thermal expansion. There are three types of thermal expansion.

1. Expansion of solids
2. Expansion of liquids
3. Expansion of gases

Expansion of Solids

Three types of expansion -takes place in solid.

Linear Expansion Expansion in length on heating is called linear expansion.

Increase in length

$l_2 = l_1(1 + \alpha\ \Delta t)$

where, l_l and l_2 are initial and final lengths,Δt = change in temperature and α = coefficient of linear expansion.

Coefficient of linear expansion

$\alpha = (\Delta l/l * \Delta t)$

where 1= real length and Δl = change in length and

Δt= change in temperature.

Superficial Expansion Expansion in area on heating is called superficial expansion.

Increase in area $A_2 = A_1(1 + \beta\ \Delta t)$

where, A_1 and A_2 are initial and final areas and β is a coefficient of superficial expansion.

Coefficient of superficial expansion

$\beta = (\Delta A/A * \Delta t)$

where. A = area, AA = change in area and At = change in temperature.

Cubical Expansion Expansion in volume on heating is called cubical expansion.

Increase in volume $V_2 = V_1(1 + \gamma\Delta t)$

where V_1 and V_2 are initial and final volumes and γ is a coefficient of cubical expansion.

Coefficient of cubical expansion

$$\gamma = \frac{\Delta V}{V \times \Delta t}$$

where V = real volume, AV =change in volume and Δt = change in temperature.
Relation between coefficients of linear, superficial and cubical expansions
$\beta = 2\alpha$ and $\gamma = 3\alpha$
Or $\alpha:\beta:\gamma = 1:2:3$

2. Expansion of Liquids

In liquids only expansion in volume takes place on heating.

(i) Apparent Expansion of Liquids When expansion of th container containing liquid, on heating is not taken into accoun then observed expansion is called apparent expansion of liquids.

Coefficient of apparent expansion of a liquid

$$(\gamma_a) = \frac{\text{apparent increase in volume}}{\text{original volume} \times \text{rise in temperature}}$$

(ii) Real Expansion of Liquids When expansion of the container, containing liquid, on heating is also taken into account, then observed expansion is called real expansion of liquids.

Coefficient of real expansion of a liquid

$$(\gamma_r) = \frac{\text{real increase in volume}}{\text{original volume} \times \text{rise in temperature}}$$

Both, y_r, and y_a are measured in $°C^{-1}$.

We can show that $y_r = y_a + y_g$

where, y_r, and y_a are coefficient of real and apparent expansion of liquids and y_g is coefficient of cubical expansion of the container.

Anamalous Expansion of Water

When temperature of water is increased from 0°C, then its vol decreases upto 4°C, becomes minimum at 4°C and then increases. behaviour of water around 4°C is called, anamalous expansion water.

3. Expansion of Gases

There are two types of coefficient of expansion in gases

(i) Volume Coefficient (γv) At constant pressure, the change in volume per unit volume per degree celsius is called volume coefficient.

$$\gamma_V = \frac{V_2 - V_1}{V_0 (t_2 - t_1)}$$

where V_0, V_1, and V_2 are volumes of the gas at 0°C, t_1°C and t_2°C.

(ii) Pressure Coefficient (γ_p) At constant volume, the change in pressure per unit pressure per degree celsius is called pressure coefficient.

$$\gamma_p = \frac{p_2 - p_1}{p_0 (t_2 - t_1)}$$

where p_0, p_1 and p_2 are pressure of the gas at 0°C, t_1° C and t_2° C.

Practical Applications of Expansion

1. When rails are laid down on the ground, space is left between the end of two rails.
2. The transmission cables are not tightly fixed to the poles.
3. The iron rim to be put on a cart wheel is always of slightly smaller diameter than that of wheel.
4. A glass stopper jammed in the neck of a glass bottle can be taken out by warming the neck of the bottles.

Important Points

- Due to increment in its time period a pendulum clock becomes slow in summer and will lose time.
- Loss of time in a time period $\Delta T = (1/2)\alpha \, \Delta\theta T$
  ∴ Loss of time in any given time interval t can be given by
  $\Delta T = (1/2)\alpha \, \Delta\theta t$
- At some higher temperature a scale will expand and scale reading will be lesser than true values, so that
  true value = scale reading $(1 + \alpha \, \Delta t)$
  Here, Δt is the temperature difference.
- However, at lower temperature scale reading will be more or true value will be less.

The branch of physics which deals with the study of transformation of heat energy into other forms of energy and vice-versa.

A thermodynamical system is said to be in thermal equilibrium when macroscopic variables (like pressure, volume, temperature, mass, composition etc) that characterise the system do not change with time.

Thermodynamical System

An assembly of an extremely large number of particles whose state can be expressed in terms of pressure, volume and temperature, is called thermodynamic system.

Thermodynamic system is classified into the following three systems

(i) **Open System** It exchange both energy and matter with surrounding.

(ii) **Closed System** It exchanges only energy (not matter) with surroundings.

(iii) **Isolated System** It exchanges neither energy nor matter with the surrounding.

A thermodynamic system is not always in equilibrium. For example, a gas allowed to expand freely against vacuum. Similary, a mixture of petrol vapour and air, when ignited by a spark is not an equilibrium state. Equilibrium is acquired eventually with time.

Thermodynamic Parameters or Coordinates or Variables

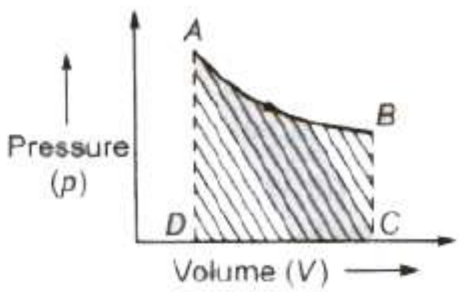
The state of thermodynamic system can be described by specifying pressure, volume, temperature, internal energy and number of moles, etc. These are called thermodynamic parameters or coordinates or variables.

Work done by a thermodynamic system is given by

$W = p * \Delta V$

where p = pressure and ΔV = change in volume.

Work done by a thermodynamic system is equal to the area enclosed between the p-V curve and the volume axis



Work done in process A-B = area ABCDA

Work done by a thermodynamic system depends not only upon the initial and final states of the system but also depend upon the path followed in the process.

Work done by the Thermodynamic System is taken as

Positive → 4 as volume increases.

Negative → 4 as volume decreases.

Internal Energy (U)

The total energy possessed by any system due to molecular motion and molecular configuration, is called its internal energy.

Internal energy of a thermodynamic system depends on temperature. It is the characteristic property of the state of the system.

Zeroth Law of Thermodynamics

According to this law, two systems in thermal equilibrium with a third system separately are in thermal equilibrium with each other. Thus, if A and B are separately in equilibrium with C, that is if $T_A = T_C$ and $T_B = T_C$, then this implies that $T_A = T_B$ i.e., the systems A and B are also in thermal equilibrium.

First Law of Thermodynamics

Heat given to a thermodynamic system (ΔQ) is partially utilized in doing work (ΔW) against the surrounding and the remaining part increases the internal energy (ΔU) of the system.

Therefore, $\Delta Q = \Delta U + \Delta W$

First law of thermodynamics is a restatement of the principle conservation of energy.

In isothermal process, change in internal energy is zero ($\Delta U = 0$).

Therefore, $\Delta Q = \Delta W$

In adiabatic process, no exchange of heat takes place, i.e., $\Delta\theta = O$.

Therefore, $\Delta U = -\Delta W$

In adiabatic process, if gas expands, its internal energy and hence, temperature decreases and vice-versa.

In isochoric process, work done is zero, i.e., $\Delta W = 0$, therefore

$\Delta Q = \Delta U$

Thermodynamic Processes

A thermodynamical process is said to take place when some changes' occur in the state of a thermodynamic system i.e., the therrnodynamie parameters of the system change with time.

(i) Isothermal Process A process taking place in a thermodynamic system at constant temperature is called an isothermal process.

Isothermal processes are very slow processes.

These process follows Boyle's law, according to which

pV = constant

From $dU = nC_v dT$ as $dT = 0$ so $dU = 0$, i.e., internal energy is constant.

From first law of thermodynamic $dQ = dW$, i.e., heat given to the system is equal to the work done by system surroundings.

Work done $W = 2.3026 \mu RT\ l0g_{10}(V_f / V_i) = 2.3026 \mu RT\ l0g_{10}(p_i / p_f)$

where, μ = number of moles, R = ideal gas constant, T = absolute temperature and V_i V_f and P_i, P_f are initial volumes and pressures.

After differentiating P V = constant, we have

$$\frac{dp}{dV} = -\frac{p}{V} \text{ and } -\frac{dp}{\frac{dV}{V}} = p$$

i.e., bulk modulus of gas in isothermal process, $\beta = p$.

P – V curve for this persons is a rectangular hyperbola

Examples

(a) Melting process is an isothermal change, because temperature of a substance remains constant during melting.

(b) Boiling process is also an isothermal operation.

(ii) Adiabatic Process A process taking place in a thermodynamic system for which there is no exchange of heat between the system and its surroundings.

Adiabatic processes are very fast processes.

These process follows Poisson's law, according to which

$$pV^{\gamma} = TV^{\gamma-1} = \frac{T^{\gamma}}{p^{\gamma-1}} = \text{constant}$$

From dQ = nCdT, C_{adi} = 0 as dQ = 0, i.e., molar heat capacity for adiabatic process is zero.

From first law, dU = – dW, i.e., work done by the system is equal to decrease in internal energy. When a system expands adiabatically, work done is positive and hence internal energy decrease, i.e., the system cools down and vice-versa.

Work done in an adiabatic process is

$$W = \frac{nR(T_i - T_f)}{\gamma - 1} = \frac{p_iV_i - p_fV_f}{\gamma - 1}$$

where T_i and T_f are initial and final temperatures. Examples

(a) Sudden compression or expansion of a gas in a container with perfectly non-conducting wall.

(b) Sudden bursting of the tube of a bicycle tyre.

(c) Propagation of sound waves in air and other gases.

(iii) Isobaric Process A process taking place in a thermodynamic system at constant pressure is called an isobaric process.

Molar heat capacity of the process is C_p and dQ = nC_pdT.

Internal energy dU = nC_v dT

From the first law of thermodynamics
dQ = dU + dW
dW = pdV = nRdT

Process equation is V / T = constant.

p- V curve is a straight line parallel to volume axis.

(iv) **Isochoric Process** A process taking place in a tlaermodynars system at constant volume is called an isochoric process.

$dQ = nC_v dT$, molar heat capacity for isochoric process is C_v.

Volume is constant, so $dW = 0$,

Process equation is p / T = constant

p- V curve is a straight line parallel to pressure axis.

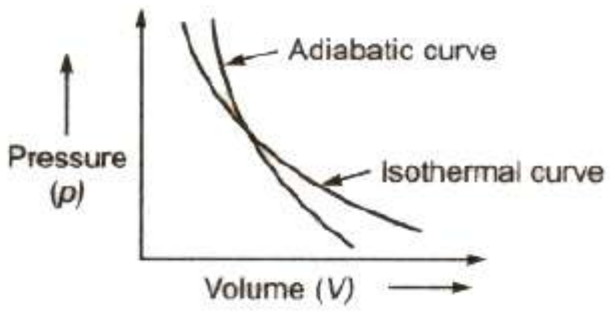
(v) **Cyclic Process** When a thermodynamic system returns to . initial state after passing through several states, then it is called cyclic process.
Efficiency of the cycle is given by

$$\eta = \frac{\text{Work done}}{\text{Heat supplied}}$$

Work done by the cycle can be computed from area enclosed cycle on p- V curve.

Isothermal and Adiabatic Curves

The graph drawn between the pressure p and the volume V of a given mass of a gas for an isothermal process is called **isothermal curve** and for an adiabatic process it is called **adiabatic curve** .



The slope of the adiabatic curve

= γ x the slope of the isothermal curve

Volume Elasticities of Gases

There are two types of volume elasticities of gases

(i) Isothermal modulus of elasticity $E_S = p$

(ii) Adiabatic modulus of elasticity $E_T = \gamma$ p

Ratio between isothermal and adiabatic modulus

$E_S / E_T = \gamma = C_p / C_V$

where C_p and C_v are specific heats of gas at constant pressure and at constant volume.

For an isothermal process $\Delta t = 0$, therefore specific heat,

c = Δ θ / m Δt = &infi;

For an adiabatic process 119= 0, therefore specific heat,

c = 0 / m Δt = 0

Second Law of Thermodynamics

The second law of thermodynamics gives a fundamental limitation to the efficiency of a heat engine and the coefficient of performance of a refrigerator. It says that efficiency of a heat engine can never be unity (or 100%). This implies that heat released to the cold reservoir can never be made zero.

Kelvin's Statement

It is impossible to obtain a continuous supply of work from a body by cooling it to a temperature below the coldest of its surroundings.

Clausius' Statement

It is impossible to transfer heat from a lower temperature body to a higher temperature body without use of an extemal agency.

Planck's Statement

It is impossible to construct a heat engine that will convert heat completely into work.

All these statements are equivalent as one can be obtained from the other.

Entropy

Entropy is a physical quantity that remains constant during a reversible adiabatic change.

Change in entropy is given by $dS = \delta Q / T$

Where, δQ = heat supplied to the system and T = absolute temperature.

Entropy of a system never decreases, i.e., dS ≥ o.

Entropy of a system increases in an irreversible process

Heat Engine

A heat energy engine is a device which converts heat energy into mechanical energy.

A heat engine consists of three parts

(i) Source of heat at higher temperature

(ii) Working substance

(iii) Sink of heat at lower temperature

Thermal efficiency of a heat engine is given by

$$\eta = \frac{\text{Work done / cycle}}{\text{Total amount of heat absorbed / cycle}}$$

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

where Q_1 is heat absorbed from the source,

Q_2 is heat rejected to the sink and T_1 and T_2 are temperatures of source and sink.

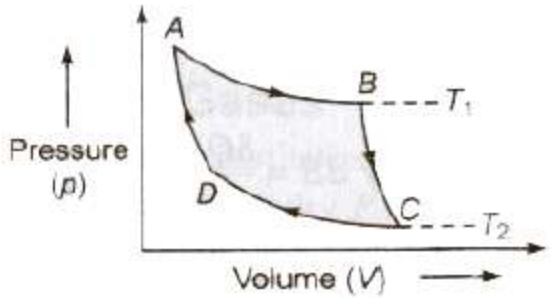
Heat engine are of two types

(i) **External Combustion Engine** In this engine fuel is burnt a chamber outside the main body of the engine. e.g., steam engine. In practical life thermal efficiency of a steam engine varies from 12% to 16%.

(ii) **Internal Combustion Engine** In this engine. fuel is burnt inside the main body of the engine. e.g., petrol and diesel engine. In practical life thermal efficiency of a petrol engine is 26% and a diesel engine is 40%.

Carnot's Cycle

Carnot devised an ideal cycle of operation for a heat engine, called Carnot's cycle.



A Carnot's cycle contains the following four processes

(i) Isothermal expansion (AB)

(ii) Adiabatic expansion (BO)

(iii) Isothermal compression (CD)

(iv) Adiabatic compression (DA)

The net work done per cycle by the engine is numerically equal to the area of the loop representing the Carnot's cycle .

After doing the calculations for different processes we can show that

$$\frac{\theta_2}{\theta_1} = \frac{T_2}{T_1}$$

Therefore, efficiency of the cycle is

$$\eta = 1 - \frac{T_2}{T_1}$$

[Efficiency of Carnot engine is maximum (not 1000/0) for given temperatures T_1 and T_2. But still Carnot engine is not a practical
engine because many ideal situations have been assumed while designing this engine which can practically not be obtained.]

Refrigerator or Heat Pump

A refrigerator or heat pump is a device used for cooling things. It absorb heat from sink at lower temperature and reject a larger amount of heat to source at higher temperature.

Coefficient of performance of refrigerator is given by

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

where Q_2 is heat absorbed from the sink, Q_1 is heat rejected to source and T_1 and T_2 are temperatures of source and sink.

Relation between efficiency (η) and coefficient of performance (β)

$$\beta = \frac{1 - \eta}{\eta}$$

Physics Notes Class 11 CHAPTER 11 THERMAL PROPERTIES OF MATTER

The branch dealing with measurement of temperature is called thremometry and the devices used to measure temperature are called thermometers.

Heat

Heat is a form of energy called thermal energy which flows from a higher temperature body to a lower temperature body when they are placed in contact.

Heat or thermal energy of a body is the sum of kinetic energies of all its constituent particles, on account of translational, vibrational and rotational motion.

The SI unit of heat energy is joule (J).

The practical unit of heat energy is calorie.

1 cal = 4.18 J

1 calorie is the quantity of heat required to raise the temperature of 1 g of water by 1°C.

Mechanical energy or work (W) can be converted into heat (Q) by 1 W = JQ

where J = Joule's mechanical equivalent of heat.

J is a conversion factor (not a physical quantity) and its value is 4.186 J/cal.

Temperature

Temperature of a body is the degree of hotness or coldness of the body. A device which is used to measure the temperature, is called a thermometer.

Highest possible temperature achieved in laboratory is about 108 while lowest possible temperature attained is 10-8 K.

Branch of Physics dealing with production and measurement temperature close to 0 K is known as cryagenics, while that deaf with the measurement of very high temperature is called pyromet Temperature of the core of the sun is 107 K while that of its surface 6000 K.

NTP or STP implies 273.15 K (0°C = 32°F).

Different Scale of Temperature

1. **Celsius Scale** In this scale of temperature, the melting point ice is taken as 0°C and the boiling point of water as 100°C and space between these two points is divided into 100 equal parts
2. **Fahrenheit Scale** In this scale of temperature, the melt point of ice is taken as 32°F and the boiling point of water as 211 and the space between these two points is divided into 180 equal parts.
3. **Kelvin Scale** In this scale of temperature, the melting pouxl ice is taken as 273 K and the boiling point of water as 373 K the space between these two points is divided into 100 equal pss

Relation between Different Scales of Temperatures

$$\frac{C}{100} = \frac{F-32}{180} = \frac{K-273}{100} = \frac{R}{80}$$

Thermometric Property

The property of an object which changes with temperature, is call thermometric property. Different thermometric properties thermometers have been given below

(i) Pressure of a Gas at Constant Volume

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

and $$p_t = p_0\left(1 + \frac{t}{273}\right)$$

$$t = \left(\frac{p_t - p_0}{p_{100} - p_0} \times 100\right) {}^\circ C$$

where p, p_{100}. and p_t, are pressure of a gas at constant volume 0°C, 100°C and t°C.

A constant volume gas thermometer can measure tempera from – 200°C to 500°C.

(ii) Electrical Resistance of Metals

$R_t = R_0(1 + \alpha t + \beta t^2)$

where α and β are constants for a metal.

As β is too small therefore we can take

$R_t = R_0(1 + \alpha t)$

where, α = temperature coefficient of resistance and
R_0 and R_t, are electrical resistances at 0°C and t°C.

$$\alpha = \frac{R_2 - R_1}{R_1 t_2 - R_2 t_1}$$

where R_1 and R_2 are electrical resistances at temperatures t_1 and t_2.

$$t = \frac{R_t - R_0}{R_{100} - R_0} \times 100°C$$

where R_{100} is the resistance at 100°C.

Platinum resistance thermometer can measure temperature from —200°C to 1200°C.

(iii) Length of Mercury Column in a Capillary Tube

$l_t = l_0(1 + \alpha t)$

where α = coefficient of linear expansion and l_0, l_t are lengths of mercury column at 0°C and t°C.

Thermo Electro Motive Force

When two junctions of a thermocouple are kept at different temperatures, then a thermo-emf is produced between the junctions, which changes with temperature difference between the junctions. Thermo-emf

$E = at + bt^2$

where a and b are constants for the pair of metals.

Unknown temperature of hot junction when cold junction is at 0°C.

$$t = \left(\frac{E_t}{E_{100}} \times 100\right)°C$$

Where E_{100} is the thermo-emf when hot junction is at 100°C.

A thermo-couple thermometer can measure temperature from —200°C to 1600°C.

Thermal Equilibrium

When there is no transfer of heat between two bodies in contact, the the bodies are called in thermal equilibrium.

Zeroth Law of Thermodynamics

If two bodies A and B are separately in thermal equilibrium with thirtli body C, then bodies A and B will be in thermal equilibrium with each other.

Triple Point of Water

The values of pressure and temperature at which water coexists inequilibrium in all three states of matter, i.e., ice, water and vapour
called triple point of water.

Triple point of water is 273 K temperature and 0.46 cm of mere pressure.

Specific Heat

The amount of heat required to raise the temperature of unit mass the substance through 1°C is called its specific heat.

It is denoted by c or s.

Its SI unit is joule/kilogram-°C'(J/kg-°C). Its dimensions is $[L^2T^{-2}\theta^{-1}]$.

The specific heat of water is 4200 J $kg^{-1}°C^{-1}$ or 1 cal g^{-1} C^{-1}, which high compared with most other substances.

Gases have two types of specific heat

1. The specific heat capacity at constant volume (C_v).
2. The specific heat capacity at constant pressure (C_r).

Specific heat at constant pressure (C_p) is greater than specific heat constant volume (C_V), i.e., $C_p > C_V$.

For molar specific heats $C_p – C_V = R$
where R = gas constant and this relation is called Mayer’s formula.

The ratio of two principal sepecific heats of a gas is represented by γ.

$$\gamma = \frac{C_p}{C_V}$$

The value of y depends on atomicity of the gas.

Amount of heat energy required to change the temperature of any substance is given by

$Q = mc\Delta t$

- where, m = mass of the substance,
- c = specific heat of the substance and
- Δt = change in temperature.

Thermal (Heat) Capacity

Heat capacity of any body is equal to the amount of heat energy required to increase its temperature through 1°C.

Heat capacity = me

where c = specific heat of the substance of the body and m = mass of the body.

Its SI unit is joule/kelvin (J/K).

Water Equivalent

It is the quantity of water whose thermal capacity is same as the heat capacity of the body. It is denoted by W.

W = ms = heat capacity of the body.

Latent Heat

The heat energy absorbed or released at constant temperature per unit mass for change of state is called latent heat.

Heat energy absorbed or released during change of state is given by

Q = mL

where m = mass of the substance and L = latent heat.

Its unit is cal/g or J/kg and its dimension is $[L^2T^{-2}]$.

For water at its normal boiling point or condensation temperature (100°C), the latent heat of vaporisation is

L = 540 cal/g
= 40.8 kJ/ mol
= 2260 kJ/kg

For water at its normal freezing temperature or melting point (0°C), the latent heat of fusion is

L = 80 cal/ g = 60 kJ/mol
= 336 kJ/kg

It is more painful to get burnt by steam rather than by boiling was 100°C gets converted to water at 100°C, then it gives out 536 heat. So, it is clear that steam at 100°C has more heat than wat 100°C (i.e., boiling of water).

After snow falls, the temperature of the atmosphere becomes very This is because the snow absorbs the heat from the atmosphere to down. So, in the mountains, when snow falls, one does not feel too but when ice melts, he feels too cold.

There is more shivering effect of ice cream on teeth as compare that of water (obtained from ice). This is because when ice cream down, it absorbs large amount of heat from teeth.

Melting

Conversion of solid into liquid state at constant temperature is melting.

Evaporation

Conversion of liquid into vapour at all temperatures (even below boiling point) is called evaporation.

Boiling

When a liquid is heated gradually, at a particular temperature saturated vapour pressure of the liquid becomes equal to atmospheric pressure, now bubbles of vapour rise to the surface d liquid. This process is called boiling of the liquid.

The temperature at which a liquid boils, is called boiling point The boiling point of water increases with increase in pre sure decreases with decrease in pressure.

Sublimation

The conversion of a solid into vapour state is called sublimation.

Hoar Frost

The conversion of vapours into solid state is called hoar fr..

Calorimetry

This is the branch of heat transfer that deals with the measorette heat. The heat is usually measured in calories or kilo calories.

Principle of Calorimetry

When a hot body is mixed with a cold body, then heat lost by ha is equal to the heat gained by cold body.

Heat lost = Heat gain

Thermal Expansion

Increase in size on heating is called thermal expansion. There are three types of thermal expansion.

1. Expansion of solids
2. Expansion of liquids
3. Expansion of gases

Expansion of Solids

Three types of expansion -takes place in solid.

Linear Expansion Expansion in length on heating is called linear expansion.

Increase in length

$l_2 = l_1(1 + \alpha\ \Delta t)$

where, l_l and l_2 are initial and final lengths,Δt = change in temperature and α = coefficient of linear expansion.

Coefficient of linear expansion

$\alpha = (\Delta l/l * \Delta t)$

where 1= real length and Δl = change in length and

Δt= change in temperature.

Superficial Expansion Expansion in area on heating is called superficial expansion.

Increase in area $A_2 = A_1(1 + \beta\ \Delta t)$

where, A_1 and A_2 are initial and final areas and β is a coefficient of superficial expansion.

Coefficient of superficial expansion

$\beta = (\Delta A/A * \Delta t)$

where. A = area, AA = change in area and At = change in temperature.

Cubical Expansion Expansion in volume on heating is called cubical expansion.

Increase in volume $V_2 = V_1(1 + \gamma\Delta t)$

where V_1 and V_2 are initial and final volumes and γ is a coefficient of cubical expansion.

Coefficient of cubical expansion

$$\gamma = \frac{\Delta V}{V \times \Delta t}$$

where V = real volume, AV =change in volume and Δt = change in temperature.
Relation between coefficients of linear, superficial and cubical expansions
$\beta = 2\alpha$ and $\gamma = 3\alpha$
Or $\alpha:\beta:\gamma = 1:2:3$

2. Expansion of Liquids

In liquids only expansion in volume takes place on heating.

(i) Apparent Expansion of Liquids When expansion of th container containing liquid, on heating is not taken into accoun then observed expansion is called apparent expansion of liquids.

Coefficient of apparent expansion of a liquid

$$(\gamma_a) = \frac{\text{apparent increase in volume}}{\text{original volume } \times \text{rise in temperature}}$$

(ii) Real Expansion of Liquids When expansion of the container, containing liquid, on heating is also taken into account, then observed expansion is called real expansion of liquids.

Coefficient of real expansion of a liquid

$$(\gamma_r) = \frac{\text{real increase in volume}}{\text{original volume } \times \text{rise in temperature}}$$

Both, y_r, and y_a are measured in °C^{-1}.

We can show that $y_r = y_a + y_g$

where, y_r, and y_a are coefficient of real and apparent expansion of liquids and y_g is coefficient of cubical expansion of the container.

Anamalous Expansion of Water

When temperature of water is increased from 0°C, then its vol decreases upto 4°C, becomes minimum at 4°C and then increases. behaviour of water around 4°C is called, anamalous expansion water.

3. Expansion of Gases

There are two types of coefficient of expansion in gases

(i) Volume Coefficient (γv) At constant pressure, the change in volume per unit volume per degree celsius is called volume coefficient.

$$\gamma_V = \frac{V_2 - V_1}{V_0 (t_2 - t_1)}$$

where V_0, V_1, and V_2 are volumes of the gas at 0°C, t_1°C and t_2°C.

(ii) Pressure Coefficient (γ_p) At constant volume, the change in pressure per unit pressure per degree celsius is called pressure coefficient.

$$\gamma_p = \frac{p_2 - p_1}{p_0 (t_2 - t_1)}$$

where p_0, p_1 and p_2 are pressure of the gas at 0°C, t_1° C and t_2° C.

Practical Applications of Expansion

1. When rails are laid down on the ground, space is left between the end of two rails.
2. The transmission cables are not tightly fixed to the poles.
3. The iron rim to be put on a cart wheel is always of slightly smaller diameter than that of wheel.
4. A glass stopper jammed in the neck of a glass bottle can be taken out by warming the neck of the bottles.

Important Points

- Due to increment in its time period a pendulum clock becomes slow in summer and will lose time.
- Loss of time in a time period $\Delta T = (1/2)\alpha\, \Delta\theta T$
  ∴ Loss of time in any given time interval t can be given by
  $\Delta T = (1/2)\alpha\, \Delta\theta t$
- At some higher temperature a scale will expand and scale reading will be lesser than true values, so that
  true value = scale reading $(1 + \alpha\, \Delta t)$
  Here, Δt is the temperature difference.
- However, at lower temperature scale reading will be more or true value will be less.

Physics Notes Class 11 CHAPTER MECHANICAL PROPERTIES OF FLUIDS part 2

Viscosity

The property of a fluid by virtue of which an internal frictional force acts between its different layers which opposes their relative motion is called viscosity.

These internal frictional force is called viscous force.

Viscous forces are intermolecular forces acting between the molecules of different layers of liquid moving with different velocities.

$$\text{Viscous force } (F) = -\eta A \frac{dv}{dx}$$

$$\eta = -\frac{F}{A\left(\frac{dv}{dx}\right)}$$

where, (dv/dx) = rate of change of velocity with distance called velocity gradient, A = area of cross-section and = coefficient of viscosity.

SI unit of η is Nsm^{-2} or pascal-second or decapoise. Its dimensional formula is $[ML^{-1}T^{-1}]$.

The knowledge of the coefficient of viscosity of different oils and its variation with temperature helps us to select a suitable lubricant for a given machine.

Viscosity is due to transport of momentum. The value of viscosity (and compressibility) for ideal liquid is zero.

The viscosity of air and of some liquids is utilised for damping the n.ving parts of some instruments.

The knowledge of viscosity of some organic liquids is used in determining the molecular weight and shape of large organic moleculars like proteins and cellulose.

Variation of Viscosity

The viscosity of liquids decreases with increase in temperature

$$\eta_t = \frac{\eta_0}{(1 + \alpha t + \beta t^2)}$$

where, η_0 and η_t is are coefficient of viscosities at 0°C t°C, α and β are constants.

The viscosity of gases increases with increase in temperatures as

$\eta \propto \sqrt{T}$

The viscosity of liquids increases with increase in pressure but the viscosity of water decreases with increase in pressure.

The viscosity of gases do not changes with pressure.

Poiseuille's Formula

The rate of flow (v) of liquid through a horizontal pipe for steady flow is given by

$$v = \frac{\pi}{8} \frac{pr^4}{\eta l}$$

where, p = pressure difference across the two ends of the tube. r = radius of the tube, n = coefficient of viscosity and 1 = length of th tube.

The Rate of Flow of Liquid

Rate of flow of liquid through a tube is given by

v = (P/R)

where, $R = (8\ \eta l/\pi r^4)$, called liquid resistance and p = liquid pressure.

(i) When two tubes are connected in series

- Resultant pressure difference $p = p_1 + p_2$
- Rate of flow of liquid (v) is same through both tubes.
- Equivalent liquid resistance, $R = R_1 + R_2$

(ii) When two tubes are connected in parallel

1. Pressure difference (p) is same across both tubes.
2. Rate of flow of liquid $v = v_1 + v_2$
3. Equivalent liquid resistance $(1/R) = (1/R_1) + (1/R_2)$

Stoke’s Law

When a small spherical body falls in a long liquid column, then after sometime it falls with a constant velocity, called terminal velocity. When a small spherical body falls in a liquid column with terminal velocity then viscous force acting on it is

$F = 6\pi\eta rv$

where, r = radius of the body, V = terminal velocity and η = coefficient of viscosity.

This is called **Stoke’s law.**

Terminal velocity $v = \frac{2}{9}\frac{r^2(\rho - \sigma)g}{\eta}$

where,

- ρ = density of body,
- σ = density of liquid,
- η = coefficient of viscosity of liquid and,
- g = acceleration due to gravity

1. If $\rho > \rho_0$, the body falls downwards.
2. If $\rho < \rho_0$, the body moves upwards with the constant velocity.
3. If ρo << ρ, $v = (2r^2\rho g/9\eta)$

Importance of Stoke’s Law

1. This law is used in the determination of electronic charge by Millikan in his oil drop experiment.
2. This law helps a man coming down with the help of parachute.
3. This law account for the formation of clouds.

Flow of Liquid

1. **Streamline Flow** The flow of liquid in which each of its particle follows the same path as followed by the proceeding particles, is called streamline flow.
2. **Laminar Flow** The steady flow of liquid over a horizontal surface in the form of layers of different velocities, is called laminar flow.
3. **Turbulent Flow** The flow of liquid with a velocity greater than its critical velocity is disordered and called turbulent flow.

Critical Velocity

The critical velocity is that velocity of liquid flow, below which its fl is streamlined and above which it becomes turbulent.

Critical velocity $v_c = (k_\eta/r\rho)$

where,

- K = Reynold's number,
- η = coefficient of viscosity of liquid
- r = radius of capillary tube and ρ = density of the liquid.

Reynold's Number

Reynold's number is a pure number and it is equal to the ratio of inertial force per unit area to the viscous force per unit area for flowing fluid.

$$\text{Reynold number } K = \frac{v_c \rho r}{\eta}$$

where, p = density of the liquid and v_c = critical velocity.

For pure water flowing in a cylindrical pipe, K is about 1000.

When 0< K< 2000, the flow of liquid is streamlined.

When 2000 < K < 3000, the flow of liquid is variable betw streamlined and turbulent.

When K > 3000, the flow of liquid is turbulent.

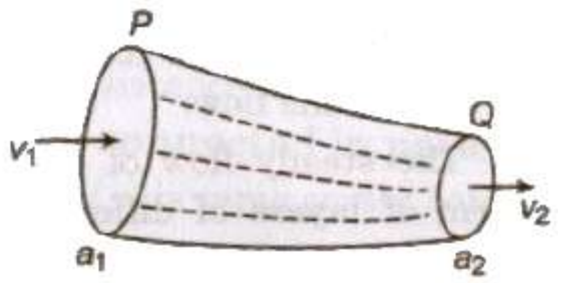
It has no unit and dimension.

Equation of Continuity

If a liquid is flowing in streamline flow in a pipe of non-unif cross-section area, then rate of flow of liquid across any cross-sec remains constant.

$a_1v_1 = a_2v_2$ av = constant

The velocity of liquid is slower where area of cross-section is larger faster where area of cross-section is smaller.



The falling stream of water becomes narrower, as the velocity of f stream of water increases and therefore its area of cross-s decreases.

Energy of a Liquid

A liquid in motion possess three types of energy

(i) Pressure Energy Pressure energy per unit mass = p/ρ

where,

p= pressure of the liquid and p = density of the liquid.
Pressure energy per unit volume = p

(ii) Kinetic Energy

- Kinetic energy per unit mass = $(1/2v^2)$
- Kinetic energy per unit volume = $1/2\rho v^2$

(iii) Potential Energy

- Potential energy per unit mass = gh
- Potential energy per unit volume = ρgh

Bernoulli's Theorem

If an ideal liquid is flowing in streamlined flow then total energy, i.e., sum of pressure energy, kinetic energy and potential energy per unit
volume of the liquid remains constant at every cross-section of the tube.

Mathematically

$$p + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

It can be expressed as

$$\frac{p}{\rho g} + \frac{v^2}{2g} + h = \text{constant}$$

where, $(p/\rho g)$ = pressure head, $(v^2/2g)$ = velocity head and h = gravitational head.

For horizontal flow of liquid,

$$p + \frac{1}{2}\rho v^2 = \text{constant}$$

- where, pis called static pressure and $(1/2\ \rho v^2)$ is called dynamic pressure.
- The erefore in horizontal flow of liquid, if p increases, v decreases and Nce-versa.
- The theorem is applicable to ideal liquid, i.e., a liquid which is non-viscous incompressible and irrotational.

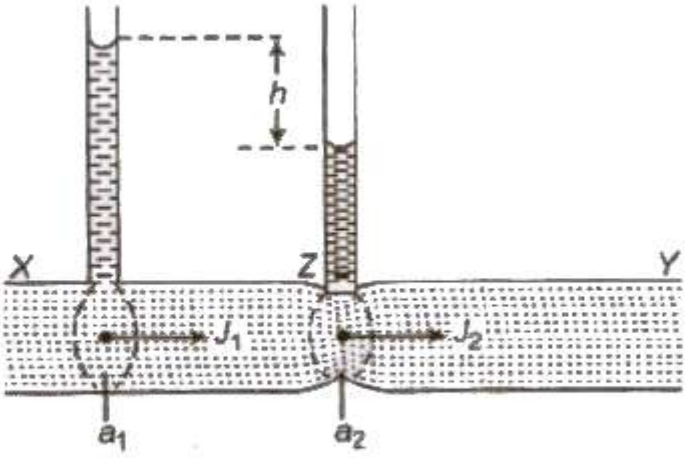
Applications of Bernoulli's Theorem

1. The action of carburetor, paintgun, scent sprayer atomiser insect sprayer is based on Bernoulli's theorem.
2. The action of Bunsen's burner, gas burner, oil stove exhaust pump is also based on Bernoulli's theorem.
3. Motion of a spinning ball (Magnus effect) is based on Bernoulli theorem.
4. Blowing of roofs by wind storms, attraction between two close parallel moving boats, fluttering of a flag etc are also based Bernoulli's theorem.

Venturimeter

It is a device used for measuring the rate of flow of liquid t pipes. Its working is based on Bernoulli's theorem.

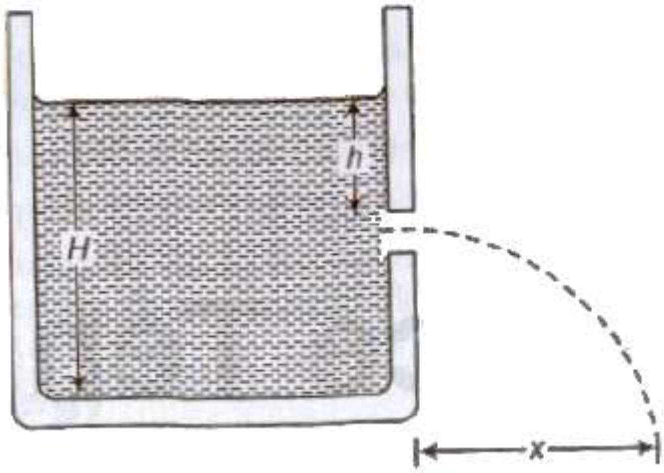
Rate of flow of liquid, $v = a_1 a_2 \sqrt{\frac{2gh}{a_1^2 - a_2^2}}$



where, a_1 and a_2 are area of cross-sections of tube at bra and narrower part and h is difference of liquid columns in ver tubes.

Torricelli's Theorem

Velocity of efflux (the velocity with which the liquid flows out orifice or narrow hole) is equal to the velocity acquired by a falling body through the same vertical distance equal to the dep orifice below the free surface of liquid.



Velocity of efflux, $v = \sqrt{2gh}$

where, h = depth of orifice below the free surface of liquid.

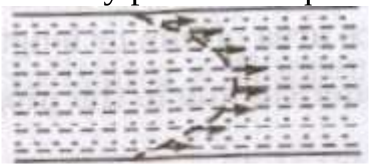
Horizontal range, $S = \sqrt{4h(H - h)}$

where, H = height of liquid column.

Horizontal range is maximum, equal to height of the liquid column H, when orifice is at half of the height of liquid column.

Important Points

- In a pipe the inner layer .central layer) have maximum velocity and the layer in contact with pipe have least velocity.
- Velocity profile of liquid flow in a pipe is parabolic.



- Solid friction is independent of area of surfaces in contact while viscous force depends on area of liquid layers.
- A lubricant is chosen ac:ording to the nature of machinary. In heavy machines lubricating oils of high viscosity are used and in light machines low viscosity oils are used.
- The cause of viscosity 1 liquids is the cohesive forces among their molecules while cause of viscosity in gases is diffusion.

Physics Notes Class 11 CHAPTER 9 MECHANICAL PROPERTIES OF SOLIDS

Deforming Force

A force which produces a change in configuration of the object on applying it, is called a deforming force.

Elasticity

Elasticity is that property of the object by virtue of which it regain its original configuration after the removal of the deforming force.

Elastic Limit

Elastic limit is the upper limit of deforming force upto which, if deforming force is removed, the body regains its original form completely and beyond which if deforming force is increased the body loses its property of elasticity and get permanently deformed.

Perfectly Elastic Bodies

Those bodies which regain its original configuration immediately and completely after the removal of deforming force are called perfectly elastic bodies. e.g., quartz and phosphor bronze etc.

Perfectly Plastic Bodies

Those bodies which does not regain its original configuration at all on the removal of deforming force are called perfectly plastic bodies, e.g., putty, paraffin, wax etc.

Stress

The internal restoring force acting per unit area of a deformed body is called stress.

Stress = Restoring force / Area

Its unit is N/m^2 or Pascal and dimensional formula is $[ML^{-12}T^{-2}]$.

Stress is a tensor quantity.

Stress is of Two Types

(i) Normal Stress If deforming force is applied normal to the area, then the stress is called normal stress.

If there is an increase in length, then stress is called **tensile stress**.

If there is a decrease in length, then stress is called **compression stress**.

(ii) **Tangential Stress** If deforming force is applied tangentially, then the stress is called tangential stress.

Strain

The fractional change in configuration is called strain.

Strain = Change in the configuration / Original configuration

It has no unit and it is a dimensionless quantity.

According to the change in configuration, the strain is of three types

(1) Longitudinal strain= Change in length / Original length

(2) Volumetric strain = Change in volume / Original volume

(iii) Shearing strain = Angular displacement of the plane perpendicular to the fixed surface.

Hooke's Law

Within the limit of elasticity, the stress is proportional to the strain.

Stress &infi; Strain

or Stress = E * Strain

where, E is the **modulus of elasticity** of the material of the body.

Types of Modulus of Elasticity

1. **Young's Modulus of Elasticity**

It is defined as the ratio of normal stress to the longitudinal strain Within the elastic limit.

y = Normal stress / Longitudinal strain

$y = F\Delta l / Al = Mg\ \Delta l / \pi r^2 l$

Its unit is N/m^2 or Pascal and its dimensional formula is $[ML^{-1}T^{-2}]$.

2. **Bulk Modulus of Elasticity**

It is defined as the ratio of normal stress to the volumetric strain within the elastic limit.

K = Normal stress / Volumetric strain

K = FV / A ΔV = &DElta;p V / Δ V

where, Δp = F / A = Change in pressure.

Its unit is N/m^2 or Pascal and its dimensional formula is [ML^{-1}T^{-2}].

3. **Modulus of Rigidity** (η)

It is defined as the ratio of tangential stress to the shearing strain, within the elastic limit.

η = Tangential stress / Shearing strain

Its urut is N/m^2 or Pascal and its dimensional formula is [ML^{-1}T^{-2}].

Compressibility

Compressibility of a material is the reciprocal of its bulk modulus of elasticity.

Compressibility (C) = 1 / k

Its SI unit is N^{-1}m^2 and CGS unit is dyne^{-1} cm^2.

Steel is more elastic than rubber. Solids are more elastic and gases are least elastic.

For liquids. modulus of rigidity is zero.
Young's modulus (Y) and modulus of rigidity (η) are possessed by solid materials only.

Limit of Elasticity

The maximum value of deforming force for which elasticity is present in the body is called its limit of elasticity.

Breaking Stress

The minimum value of stress required to break a wire, is called breaking stress.

Breaking stress is fixed for a material but breaking force varies with area of cross-section of the wire.

Safety factor = Breaking stress / Working stress

Elastic Relaxation Time

The time delay in restoring the original configuration after removal of deforming force is called elastic relaxation time.

For quartz and phosphor bronze this time is negligible.

Elastic After Effect

The temporary delay in regaining the original configuration by the elastic body after the removal of deforming force, is called elastic after effect.

Elastic Fatigue

The property of an elastic body by virtue of which its behaviour becomes less elastic under the action of repeated alternating deforming force is called elastic fatigue.

Ductile Materials

The materials which show large plastic range beyond elastic limit are called ductile materials, e.g., copper, silver, iron, aluminum, etc.

Ductile materials are used for making springs and sheets.

Brittle Materials

The materials which show very small plastic range beyond elastic limit are called brittle materials, e.g., glass, cast iron, etc.

Elastomers

The materials for which strain produced is much larger than the stress applied, with in the limit of elasticity are called elastomers, e.g., rubber, the elastic tissue of aorta, the large vessel carrying blood from heart. etc.

Elastomers have no plastic range.

Elastic Potential Energy in a Stretched Wire

The work done in stretching a wire is stored in form of potential energy of the wire.

Potential energy U = Average force * Increase in length

$= 1 / 2 \, F\Delta l$

= 1 / 2 Stress * Strain * Volume of the wire

Elastic potential energy per unit volume

U = 1 / 2 * Stress * Strain

= 1 / 2 (Young's modulus) * $(\text{Strain})^2$

Elastic potential energy of a stretched spring = 1 / 2 kx^2

where, k = Force constant of spring and x = Change in length.

Thermal Stress

When temperature of a rod fixed at its both ends is changed, then the produced stress is called thermal stress.

Thermal stress = F / A = $y\alpha\Delta\theta$

where, α = coefficient of linear expansion of the material of the rod.

When temperature of a gas enclosed in a vessel is changed, then the thermal stress produced is equal to change in pressure (Δp)of the gas.

Thermal stress = Δ p = Ky Δ θ

where, K = bulk modulus of elasticity and

γ = coefficient of cubical expansion of the gas.

Interatomic force constant

$K = Yr_o$

where, r_o = interatomic distance.

Poisson's Ratio

When a deforming force is applied at the free end of a suspended wire of length 1 and radius R, then its length increases by dl but its radius decreases by dR. Now two types of strains are produced by a single force.

(i) Longitudinal strain = &DElta;U l

(ii) Lateral strain = – Δ R/ R

∴ Poisson's Ratio (σ) = Lateral strain / Longitudinal strain = – Δ R/ R / ΔU l

The theoretical value of Poisson's ratio lies between – 1 and 0.5. Its practical value lies between 0 and 0.5.

Relation Between Y, K, η and σ

(i) $Y = 3K (1 – 2\sigma)$

(ii) $Y = 2 \eta (1 + \sigma)$

(iii) $\sigma = 3K – 2\eta / 2\eta + 6K$

(iv) $9 / Y = 1 / K + 3 / \eta$ or $Y = 9K \eta / \eta + 3K$

Important Points

- Coefficient of elasticity depends upon the material, its temperature and purity but not on stress or strain.
- For the same material, the three coefficients of elasticity γ, η and K have different magnitudes.
- Isothermal elasticity of a gas $E_T = \rho$ where, ρ = pressure of the gas.
- Adiabatic elasticity of a gas $E_s = \gamma\rho$

where, $\gamma = C_p / C_v$ ratio of specific heats at constant pressure and at constant volume.

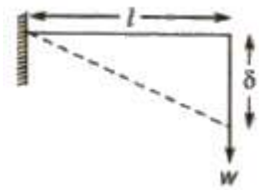
- Ratio between isothermal elasticity and adiabatic elasticity $E_s/ E_T = \gamma = C_p / C_v$

Cantilever

A beam clamped at one end and loaded at free end is called a cantilever.

Depression at the free end of a cantilever is given by

$\delta = wl^3 / 3YI_G$



where, w = load, 1 = length of the cantilever,

y = Young’s modulus of elasticity, and I_G = geometrical moment of inertia.

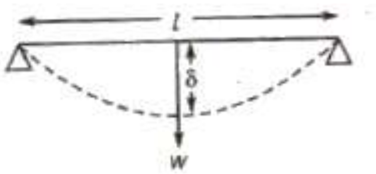
For a beam of rectangular cross-section having breadth b and thickness d.

$I_G = bd^3 / 12$

For a beam of circular cross-section area having radius r,

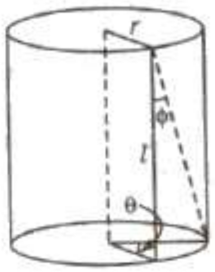
$I_G = \pi r^4 / 4$

Beam Supported at Two Ends and Loaded at the Middle



Depression at middle $\delta = wl^3 / 48YI_G$

Torsion of a Cylinder



Couple per unit twist $\quad C = \dfrac{\pi \eta r^4}{2l}$

where, η = modulus of rigidity of the material of cylinder,

r = radius of cylinder,

and l = length of cylinder,

Work done in twisting the cylinder through an angle θ

$W = 1 / 2\ C\theta^2$

Relation between angle of twist (θ) and angle of shear (φ)

$r\theta = l\varphi$ or $\varphi = r / l = \theta$

Physics Notes Class 11 CHAPTER 8 GRAVITATION

Every object in the universe attracts every other object with a force which is called the force of **gravitation.**

Gravitation is one of the four classes of interactions found in nature.

These are

(i) the gravitational force

(ii) the electromagnetic force

(iii) the strong nuclear force (also called the hadronic force).

(iv) the weak nuclear forces.

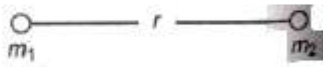
Although, of negligible importance in the interactions of elementary particles, gravity is of primary importance in the interactions of objects. It is gravity that holds the universe together.

Newton's Law of Gravitation

Gravitational force is a attractive force between two masses m_1 and m_2 separated by a distance r.

The gravitational force acting between two point objects is proportional to the product of their masses and inversely proportional to the square of the distance between them.

Gravitational force. $F = \frac{Gm_1m_2}{r^2}$



where G is universal gravitational constant.

The value of G is 6.67 X 10^{-11} Nm^2 kg^{-2} and is same throughout the universe.

The value of G is independent of the nature and size of the bodies well as the nature of the medium between them.

Dimensional formula of Gis $[M^{-1}L^3T^{-2}]$.

Important Points about Gravitation Force

(i) Gravitational force is a central as well as conservative force.

(ii) It is the weakest force in nature.

(iii) It is 1036 times smaller than electrostatic force and 10’l8times smaller than nuclear force.

(iv) The law of gravitational is applicable for all bodies, irrespective of their size, shape and position.

(v) Gravitational force acting between sun and planet provide it centripetal force for orbital motion.

(vi) Gravitational pull of the earth is called gravity.

(vii) Newton’s third law of motion holds good for the force of gravitation. It means the gravitation forces between two bodies are action-reaction pairs.

Following three points are important regarding the gravitational force

(i) Unlike the electrostatic force, it is independent of the medium between the particles.

(ii) It is conservative in nature.

(iii) It expresses the force between two point masses (of negligible volume). However, for external points of spherical bodies the whole mass can be assumed to be concentrated at its centre of mass.

Note Newton’s law of gravitation holde goods for object lying at uery large distances and also at very short distances. It fails when the distance between the objects is less than 10-9 m i.e., of the order of intermolecular distances.

Acceleration Due to Gravity

The uniform acceleration produced in a freely falling object due to the gravitational pull of the earth is known as acceleration due to gravity.

It is denoted by g and its unit is m/s^2. It is a vector quantity and its direction is towards the centre of the earth.

The value of g is independent of the mass of the object which is falling freely under gravity.

The value of g changes slightly from place to place. The value of g is taken to be 9.8 m/s^2 for all practical purposes.

The value of acceleration due to gravity on the moon is about. one sixth of that On the earth and on the sun is about 27 times of that on the earth.

Among the planets, the acceleration due to gravity is minimum on the mercury.

Relation between g and a is given by

$g = Gm / R^2$

where M = mass of the earth = $6.0 * 10^{24}$ kg and R = radius of the earth = $6.38 * 10^6$ m.

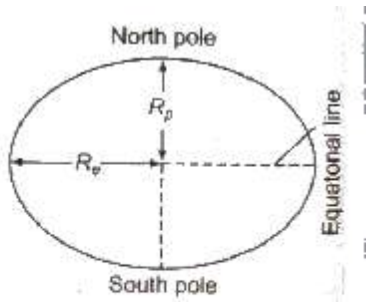
Acceleration due to gravity at a height h above the surface of the earth is given by

$g_h = Gm / (R+h)^2 = g (1 – 2h / R)$

Factors Affecting Acceleration Due to Gravity

(i) **Shape of Earth** Acceleration due to gravity g &infi; $1 / R^2$ Earth is elliptical in shape. Its diameter at poles is approximately 42 km less than its diameter at equator.

Therefore, g is minimum at equator and maximum at poles.



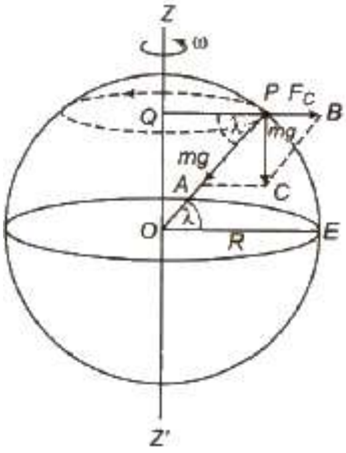
(ii) **Rotation of Earth about Its Own Axis** If ω is the angular velocity of rotation of earth about its own axis, then acceleration due to gravity at a place having latitude λ is given by

$g' = g – R\omega^2 \cos^2 \lambda$

At poles $\lambda = 90°$ and $g' = g$

Therefore, there is no effect of rotation of earth about its own axis at poles.

At equator $\lambda = 0°$ and $g' = g – R\omega^2$



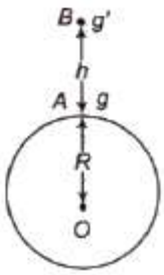
The value of g is minimum at equator

If earth stapes its rotation about its own axis, then g will remain unchanged at poles but increases by $R\omega^2$at equator.

(iii) **Effect of Altitude** The value of g at height h from earth's surface

g' = g / (1 + h / R)^2

Therefore g decreases with altitude.

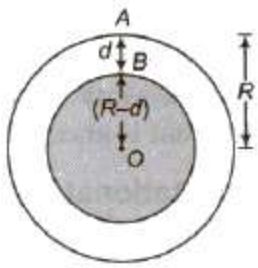


(iv) **Effect of Depth** The value of gat depth h A from earth's surface

g' = g * (1 – h / R)

Therefore g decreases with depth from earth's surface.

The value of g becomes zero at earth's centre.



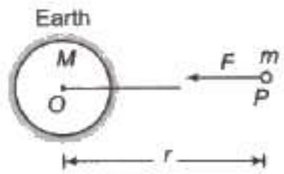
Gravitational Field

The space in the surrounding of any body in which its gravitational pull can be experienced by other bodies is called **gravitational field**.

Intensity of Gravitational Field

The gravitational force acting per unit mass at Earth any point in gravitational field is called intensity of gravitational field at that point.
It is denoted by E_g or I.

E_g or I = F / m



Intensity of gravitational field at a distance r from a body of mass M is given by

E_g or I = GM / r^2

It is a vector quantity and its direction is towards the centre of gravity of the body.

Its S1 unit is N/m and its dimensional formula is $[LT^{-2}]$.

Gravitational mass M_g is defined by Newton's law of gravitation.

$M_g = F_g / g = W / g$ = Weight of body / Acceleration due to gravity

$\therefore (M_1)g / (M_2)g = F_{g1g2} / F_{g2g1}$

Gravitational Potential

Gravitational potential at any point in gravitational field is equal the work done per unit mass in bringing a very light body from infinity to that point.

It is denoted by V_g.

Gravitational potential, $V_g = W / m = – GM / r$

Its SI unit is J / kg and it is a scalar quantity. Its dimensional formula is $[L^3r^{-2}]$.

Since work W is obtained, that is, it is negative, the gravitational potential is always negative.

Gravitational Potential Energy

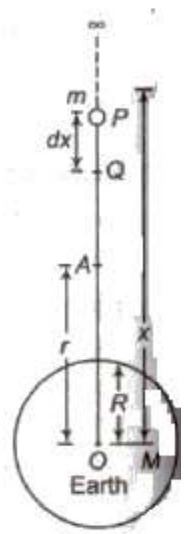
Gravitational potential energy of any object at any point in gravitational field is equal to the work done in bringing it from infinity to that point. It is denoted by U.

Gravitational potential energy $U = – GMm / r$

The negative sign shows that the gravitational potential energy decreases with increase in distance.

Gravitational potential energy at height h from surface of earth

$U_h = – GMm / R + h = mgR / 1 + h/R$



Satellite

A heavenly object which revolves around a planet is called a satellite. Natural satellites are those heavenly objects which are not man made and revolve around the earth. Artificial satellites are those neaven objects which are man made and launched for some purposes revolve around the earth.

Time period of satellite

$T = 2\pi \sqrt{r^3 / GM}$

$= 2\pi \sqrt{(R + h)^3 / g}$ [$g = GM / R^2$

Near the earth surface, time period of the satellite

$T = 2\pi \sqrt{R^3 / GM} = \sqrt{3\pi / Gp}$

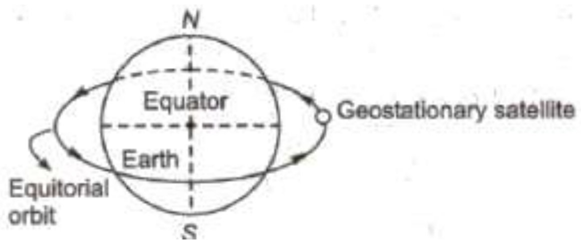
$T = 2\pi \sqrt{R / g} = 5.08 * 10^3$ s = 84 min.

where p is the average density of earth.

Artificial satellites are of two types :

1. **Geostationary or Parking Satellites**

A satellite which appears to be at a fixed position at a definite height to an observer on earth is called geostationary or parking satellite.



Height from earth's surface = 36000 km

Radius of orbit = 42400 km

Time period = 24 h

Orbital velocity = 3.1 km/s

Angular velocity = $2\pi / 24 = \pi / 12$ rad / h

There satellites revolve around the earth in equatorial orbits.

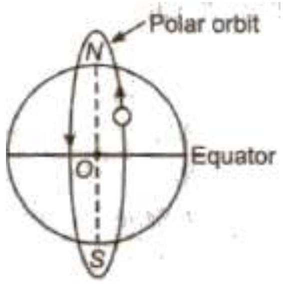
The angular velocity of the satellite is same in magnitude and direction as that of angular velocity of the earth about its own axis.

These satellites are used in communication purpose.

INSAT 2B and INSAT 2C are geostationary satellites of India.

2. **Polar Satellites**

These are those satellites which revolve in polar orbits around earth. A polar orbit is that orbit whose angle of inclination with equatorial plane of earth is 90°.



Height from earth's surface = 880 km

Time period = 84 min

Orbital velocity = 8 km / s

Angular velocity = $2\pi / 84 = \pi / 42$ rad / min.

There satellites revolve around the earth in polar orbits.

These satellites are used in forecasting weather, studying the upper region of the atmosphere, in mapping, etc.

PSLV series satellites are polar satellites of India.

Orbital Velocity

Orbital velocity of a satellite is the minimum velocity required to the satellite into a given orbit around earth.

Orbital velocity of a satellite is given by

$v_o = \sqrt{GM / r} = R \sqrt{g / R + h}$

where, M = mass of the planet, R = radius of the planet and h = height of the satellite from planet's surface.

If satellite is revolving near the earth's surface, then r = (R + h) =- R

Now orbital velocity,

$v_o = \sqrt{gR}$

= 7.92km / h

if v is the speed of a satellite in its orbit and v_o is the required orbital velocity to move in the orbit, then

(i) If $v < v_o$, then satellite will move on a parabolic path and satellite falls back to earth.

(ii) If $V = v_o$ then satellite revolves in circular path/orbit around earth.

(iii) If $v_o < V < v_e$ then satellite shall revolve around earth in elliptical orbit.

Energy of a Satellite in Orbit

Total energy of a satellite

E = KE + PE

= GMm / 2r + (- GMm / r)

= – GMm / 2r

Binding Energy

The energy required to remove a satellite from its orbit around the earth (planet) to infinity is called binding energy of the satellite.

Binding energy of the satellite of mass m is given by

BE = + GMm / 2r

Escape Velocity

Escape velocity on earth is the minimum velocity with which a body has to be projected vertically upwards from the earth's surface so that it just crosses the earth's gravitational field and never returns.

Escape velocity of any object

$v_e = \sqrt{2}GM / R$

$= \sqrt{2}gR = \sqrt{8}\pi p\ GR^2 / 3$

Escape velocity does not depend upon the mass or shape or size of the body as well as the direction of projection of the body.

Escape velocity at earth is 11.2 km / s.

Some Important Escape Velocities

Heavenly body	Escape velocity
Moon	2.3 km/s

Mercury	4.28 km/s
Earth	11.2 km/s
Jupiter	60 km/s
Sun	618 km/s
Neutron star	2×10^5 km/s

Relation between escape velocity and orbital velocity of the satellite

$v_e = \sqrt{2}\, v_o$

If velocity of projection U is equal the escape velocity ($v = v_e$), then the satellite will escape away following a parabolic path.

If velocity of projection u of satellite is greater than the escape velocity ($v > v_e$), then the satellite will escape away following a hyperbolic path.

Weightlessness

It is a situation in which the effective weight of the body becomes zero,

Weightlessness is achieved

(i) during freely falling under gravity

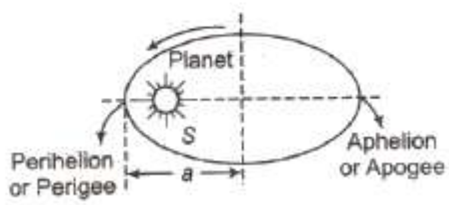
(ii) inside a space craft or satellite

(iii) at the centre of the earth

(iv) when a body is lying in a freely falling lift.

Kepler's Laws of Planetary Motion

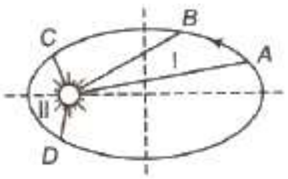
(i) **Law of orbit** Every planet revolve around the sun in elliptical orbit and sun is at its one focus.



(ii) **Law of area** The radius vector drawn from the sun to a planet sweeps out equal areas in equal intervals of time, i.e., the areal velocity of the planet around the sun is constant.

Areal velocity of a planet

$dA / dt = L / 2m$ = constant



where L = angular momentum and m = mass of the planet.

(iii) **Law of period** The square of the time period of revolution of planet around the sun is directly proportional to the cube semi-major axis of its elliptical orbit.

T^2 &infi; a^3 or $(T_1 / T_2)^2 = (a_1 / a_2)^3$

where, a = semi-major axis of the elliptical orbit.

Important Points

(i) A missile is launched with a velocity less than the escape velocity. The sum of its kinetic energy and potential energy is negative.

(ii) The orbital speed of jupiter is less than the orbital speed of earth.

(iii) A bomb explodes on the moon. You cannot hear the sound of the explosion on earth.

(iv) A bottle filled with water at 30°C and fitted with a cork is taken to the moon. If the cork is opened at the surface of the moon then water will boil.

(v) For a satellite orbiting near earth's surface

(a) Orbital velocity = 8 km / s

(b) Time period = 84 min approximately

(c) Angular speed $\omega = 2\pi / 84$ rad / min

= 0.00125 rad / s

(vi) Inertial mass and gravitational mass

(a) Inertial mass = force / acceleration

(b) Gravitational mass = weight of body / acceleration due to gravity

(c) They are equal to each other in magnitude.

(d) Gravitational mass of a body is affected by the presence of other bodies near it. Inertial mass of a body remains unaffected by the presence of other bodies near it.

Physics Notes Class 11 CHAPTER 7 SYSTEM OF PARTICLES AND ROTATIONAL MOTION

Centre of Mass

Centre of mass of a system is the point that behaves as whole mass of the system is concentrated at it and all external forces are acting on it.

For rigid bodies, centre of mass is independent of the state of the body i.e., whether it is in rest or in accelerated motion centre of mass will rermain same.

Centre of Mass of System of n Particles

If a system consists of n particles of masses m_1, m_2, m_3 ,… m_n having position vectors r_l, r_2, r_3,… r_n. then position vector of centre of mass of

the system, $$\mathbf{r}_{CM} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2 + m_3\mathbf{r}_3 + \ldots + m_n\mathbf{r}_n}{m_1 + m_2 + m_3 + \ldots + m_n} = \frac{\sum_{i=1}^{n} m_i\mathbf{r}_i}{\sum m_i}$$

In terms of coordinates,

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + \ldots + m_nx_n}{m_1 + m_2 + \ldots + m_n} = \frac{\sum_{i=1}^{n} m_i\mathbf{x}_i}{\sum m_i}$$

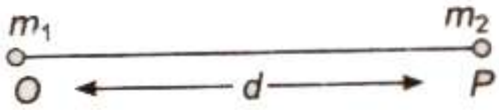
$$y_{CM} = \frac{m_1y_1 + m_2y_2 + \ldots + m_ny_n}{m_1 + m_2 + \ldots + m_n} = \frac{\sum_{i=1}^{n} m_i\mathbf{y}_i}{\sum m_i}$$

$$z_{CM} = \frac{m_1z_1 + m_2z_2 + \ldots + m_nz_n}{m_1 + m_2 + \ldots + m_n} = \frac{\sum_{i=1}^{n} m_i\mathbf{z}_i}{\sum m_i}$$

Centre of Mass of Two Particle System

Choosing O as origin of the coordinate axis.

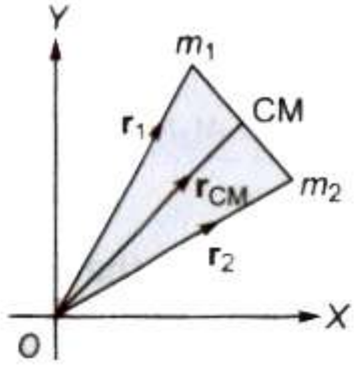
(i) Then, position of centre of mass from $m_1 = \dfrac{m_2 d}{m_1 + m_2}$



(ii) Position of centre of mass from $m_2 = (m_1 d) / m_1 + m_2$

iii) If position vectors of particles of masses m_1 and m_2 are r_1 and $r_2$respectively, then

$$\mathbf{r}_{CM} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}$$



(iv) If in a two particle system, particles of masses m_1 and m_2 moving with velocities v_1 and $v_2$respectively, then velocity the centre of mass

$$\mathbf{v}_{CM} = \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_2}{m_1 + m_2}$$

(v) If accelerations of the particles are a_1, and $a_1$respectively, then acceleration of the centre of mass

$$\mathbf{a}_{CM} = \frac{m_1\mathbf{a}_1 + m_2\mathbf{a}_2}{m_1 + m_2}$$

(vi) Centre of mass of an isolated system has a constant velocity.

(vii) It means isolated system will remain at rest if it is initially rest or will move with a same velocity if it is in motion initially.

(viii) The position of centre of mass depends upon the shape, size and distribution of the mass of the body.

(ix) The centre of mass of an object need not to lie with in the object.

(x) In symmetrical bodies having homogeneous distribution mass the centre of mass coincides with the geometrical centre the body.

(xi) The position of centre of mass of an object changes translatory motion but remains unchanged in rotatory motion,

Translational Motion

A rigid body performs a pure translational motion, if each particle the body undergoes the same displacement in the same direction in given interval of time.

Rotational Motion

A rigid body performs a pure rotational motion, if each particle of the body moves in a circle, and the centre of all the circles lie on a straight line called the axes of rotation.

Rigid Body

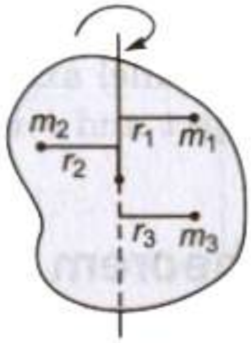
If the relative distance between the particles of a system do not changes on applying force, then it called a rigtd body. General motion of a rigid body consists of both the translational motion and the rotational motion.

Moment of Inertia

The inertia of rotational motion is called moment of inertia. It is denoted by L.

Moment of inertia is the property of an object by virtue of which it opposes any change in its state of rotation about an axis.

The moment of inertia of a body about a given axis is equal to the sum of the products of the masses of its constituent particles and the square of their respective distances from the axis of rotation.



Axis of rotation

Moment of inertia of a body

$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + \ldots = \sum_{i=1}^{n} m_ir_i^2$$

Its unit is $kg.m^2$ and its dimensional formula is $[ML^2]$.

The moment of inertia of a body depends upon

- position of the axis of rotation
- orientation of the axis of rotation
- shape and size of the body
- distribution of mass of the body about the axis of rotation.

The physical significance of the moment of inertia is same in rotational motion as the mass in linear motion.

The Radius of Gyration

The root mean square distance of its constituent particles from the axis of rotation is called the radius of gyration of a body.

It is denoted by K.

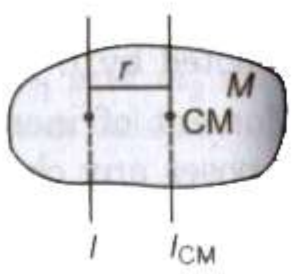
Radius of gyration

$$K = \sqrt{\frac{r_1^2 + r_2^2 + \ldots + r_n^2}{n}}$$

The product of the mass of the body (M) and square of its radius gyration (K) gives the same moment of inertia of the body about rotational axis.

Therefore, moment of inertia $I = MK^2 \Rightarrow K = \sqrt{1/M}$

Parallel Axes Theorem

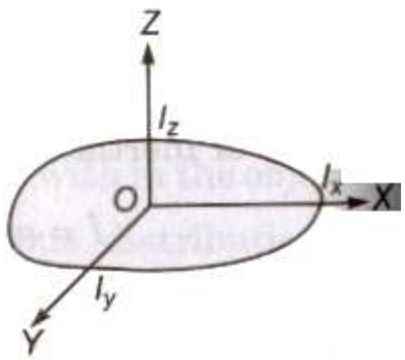


The moment of inertia of any object about any arbitrary axes is equal to the sum of moment of inertia about a parallel axis passing through the centre of mass and the product of mass of the body and the square of the perpendicular distance between
the two axes.

Mathematically $I = I_{CM} + Mr^2$

where I is the moment of inertia about the arbitrary axis, I_{cM} is moment of inertia about the parallel axis through the centre of mass, M is the total mass of the object and r is the perpendicular distance between the axis.

Perpendicular Axes Theorem



The moment of inertia of any two dimensional body about an axis perpendicular to its plane (I_z) is equal to the sum of moments of inertia of the body about two mutually perpendicular axes lying in its own plane and intersecting
each other at a point, where the perpendicular axis passes through it.

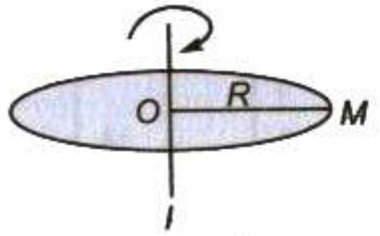
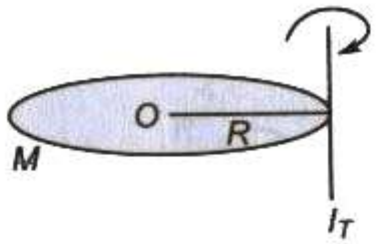
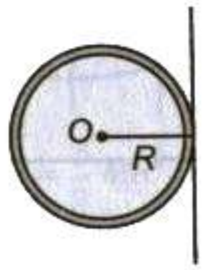
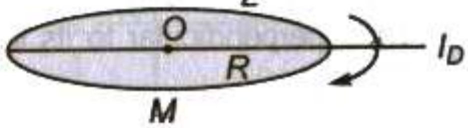
Mathematically $I_z = I_x + I_y$

where I_x and I_y are the moments of inertia of plane lamina about perpendicular axes X and Y respectively which lie in the plane lamina an intersect each other.

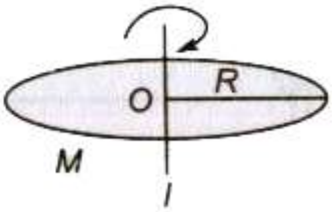
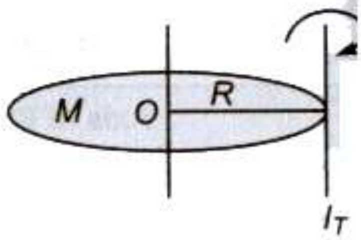
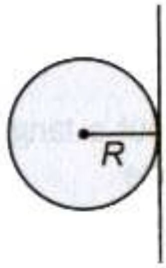
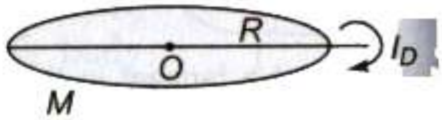
Theorem of parallel axes is applicable for any type of rigid body whether it is a two dimensional or three dimensional, while the theorem of perpendicular is applicable for laminar type or two I dimensional bodies only.

Moment of Inertia of Homogeneous Rigid Bodies

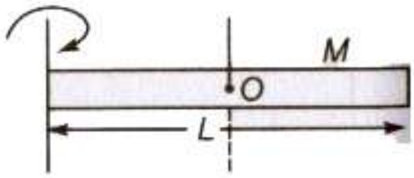
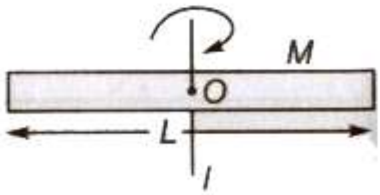
For a Thin Circular Ring

S. No.	Axis of Rotation	Moment of Inertia
(a)	About an axis passing through its centre and perpendicular to its plane	 $I = MR^2$
(b)	About a tangent perpendicular to its plane	 $I_T = 2MR^2$
(c)	About a tangent in the plane of ring	 $I_{T'} = \frac{3}{2} MR^2$
(d)	About a diameter	 $I_D = \frac{1}{2} MR^2$

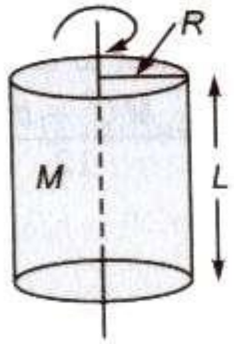
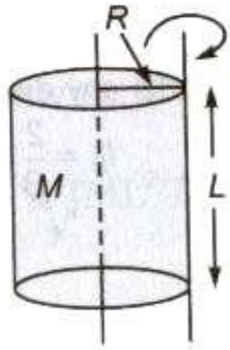
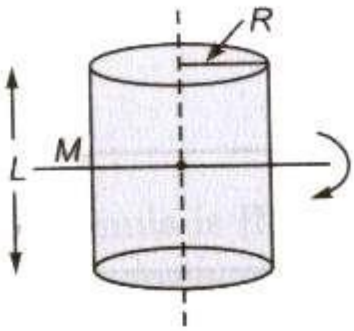
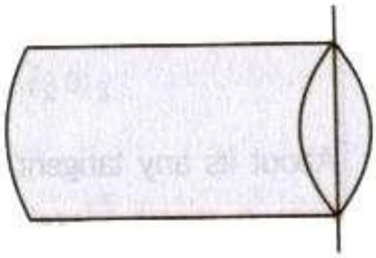
For a Circular Disc

S. No.	Axis of Rotation	Moment of Inertia
(a)	About an axis passing through its centre and perpendicular to its plane	 $I = \frac{1}{2} MR^2$
(b)	About a tangent perpendicular to its plane	 $I_T = \frac{3}{2} MR^2$
(c)	About a tangent in its plane	 $I_{T'} = \frac{5}{4} MR^2$
(d)	About a diameter	 $I_D = \frac{1}{4} MR^2$

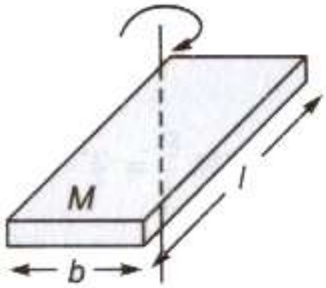
For a Thin Rod

S. No.	Axis of Rotation	Moment of Inertia
(a)	About an axis passing through its centre and perpendicular to its length	 $I = \frac{ML^2}{12}$
(b)	About an axis passing through its one end and perpendicular to its length	 $I = \frac{ML^2}{3}$

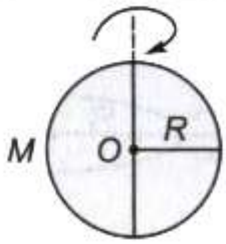
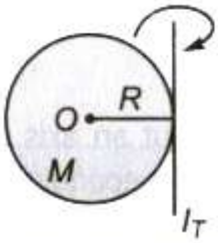
For a Solid Cylinder

S. No.	Axis of Rotation	Moment of Inertia
(a)	About its geometrical axis	 $I = \frac{MR^2}{2}$
(b)	About an axis passing through its outer face along its length	 $I = \frac{3}{2}MR^2$
(c)	About an axis passing through its centre and pependicular to its length	 $I = \left(\frac{ML^2}{12} + \frac{MR^2}{4}\right)$
(d)	About an axis passing through its diameter of circular surface.	 $I = \left(\frac{ML^2}{3} + \frac{MR^2}{4}\right)$

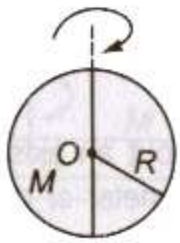
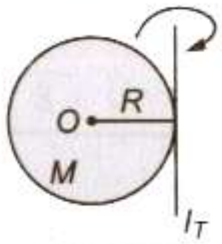
For a Rectangular Plate

Axis of Rotation	Moment of Inertia
About an axis passing through its centre and perpendicular to its plane $I = \frac{M(l^2 + b^2)}{12}$	

For a Thin Spherical Shell

S. No.	Axis of Rotation	Moment of Inertia
(a)	About its any diameter $I_D = \frac{2}{3} MR^2$	
(b)	About its any tangent $I_T = \frac{5}{3} MR^2$	

For a Solid Sphere

S. No.	Axis of Rotation	Moment of Inertia
(a)	About its any diameter $I_D = \frac{2}{5} MR^2$	
(b)	About its any tangent $I_T = \frac{7}{5} MR^2$	

Equations of Rotational Motion

(i) $\omega = \omega_0 + \alpha t$

(ii) $\theta = \omega_0 t + 1/2\ \alpha t^2$

(iii) $\omega^2 = \omega_0^2 + 2\alpha\theta$

where θ is displacement in rotational motion, ω_0 is initial velocity, omega; is final velocity and a is acceleration.

Torque

Torque or moment of a force about the axis of rotation

$\tau = r \times F = rF \sin\theta\ n$

It is a vector quantity.

If the nature of the force is to rotate the object clockwise, then torque is called negative and if rotate the object anticlockwise, then it is called positive.

Its SI unit is 'newton-metre' and its dimension is $[ML^2T^{-2}]$.

In rotational motion, torque, $\tau = I\alpha$

where a is angular acceleration and 1is moment of inertia.

Angular Momentum

The moment of linear momentum is called angular momentum.

It is denoted by L.

Angular momentum, $L = I\ \omega = mvr$

In vector form, $L = I\,\omega = r \times mv$
Its unit is 'joule-second' and its dimensional formula is $[ML^2T^{-1}]$.

Torque, $\tau = dL/dt$

Conservation of Angular Momentum

If the external torque acting on a system is zero, then its angular momentum remains conserved.
If τ_{ext} 0, then $L = I(\omega) =$ constant $\Rightarrow I_1\omega_1 == I_2\omega_2$

Angular Impulse

Total effect of a torque applied on a rotating body in a given time is called angular impulse. Angular impulse is equal to total change in angular momentum of the system in given time. Thus, angular impulse

$$J = \int_0^{\Delta L} \tau\, dt = L_f - L_i$$

Physics Notes Class 11 CHAPTER 5 LAWS OF MOTION

Inertia

The property of an object by virtue of which it cannot change its state of rest or of uniform motion along a straight line its own, is called **inertia.**

Inertia is a measure of mass of a body. Greater the mass of a body greater will be its inertia or vice-versa.

Inertia is of three types:

(i) **Inertia of Rest** When a bus or train starts to move suddenly, the passengers sitting in it falls backward due to inertia of rest.

(ii) **Inertia of Motion** When a moving bus or train stops suddenly, the passengers sitting in it jerks in forward direction due to inertia of motion.

(iii) **Inertia of Direction** We can protect yourself from rain by an umbrella because rain drops can not change its direction its own due to inertia of direction.

Force

Force is a push or pull which changes or tries to change the state of rest, the state of uniform motion, size or shape of a body.

Its SI unit is newton (N) and its dimensional formula is $[MLT^{-2}]$.

Forces can be categorized into two types:

(i) **Contact Forces** Frictional force, tensional force, spring force, normal force, etc are the contact forces.

(ii) **Action at a Distance Forces** Electrostatic force, gravitational force, magnetic force, etc are action at a distance forces.

Impulsive Force

A force which acts on body for a short interval of time, and produces a large change in momentum is called an impulsive force.

Linear Momentum

The total amount of motion present in a body is called its momentum. Linear momentum of a body is equal to the product of its mass and velocity. It is denoted by p.

Linear momentum p = mu.

Its S1 unit is kg-m/s and dimensional formula is $[MLT^{-1}]$.

It is a vector quantity and its direction is in the direction of velocity of the body.

Impulse

The product of impulsive force and time for which it acts is called impulse.

Impulse = Force * Time = Change in momentum

Its S1 unit is newton-second or kg-m/s and its dimension is $[MLT^{-1}]$.

It is a vector quantity and its direction is in the direction of force.

Newton's Laws of Motion

1. **Newton's First Law of Motion**

A body continues to be in its state of rest or in uniform motion along a straight line unless an external force is applied on it.

This law is also called **law of inertia**.

Examples

(i) When a carpet or a blanket is beaten with a stick then the dust particles separate out from it.

(ii) If a moving vehicle suddenly stops then the passengers inside the vehicle bend outward.

2. **Newton's Second Law of Motion**

The rate of change of linear momentum is proportional to the applied force and change in momentum takes place in the direction of applied force.

Mathematically F &infi; dp / dt

F = k (d / dt) (mv)

where, k is a constant of proportionality and its value is one in SI and CGS system.

F= mdv / dt = ma

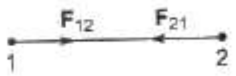
Examples

(i) It is easier for a strong adult to push a full shopping cart than it is for a baby to push the same cart. (This is depending on the net force acting on the object).

(ii) It is easier for a person to push an empty shopping cart than a full one (This is depending on the mass of the object).

3. **Newton's Third Law of Motion**

For every action there is an equal and opposite reaction and both acts on two different bodies



Mathematically $F_{12} = -F_{21}$

Examples

(i) Swimming becomes possible because of third law of motion.

(ii) Jumping of a man from a boat onto the bank of a river.

(iii) Jerk is produced in a gun when bullet is fired from it.

(iv) Pulling of cart by a horse.

Note Newton's second law of motion is called real law of motion because first and third laws of motion can be obtained by it.

The modern version of these laws is

(i) A body continues in its initial state of rest or motion with uniform velocity unless acted on by an unbalanced external force.

(ii) Forces always occur in pairs. If body A exerts a force on body B, an equal but opposite force is exerted by body B on body A.

Law of Conservation of Linear Momentum

If no external force acts on a system, then its total linear momentum remains conserved.

Linear momentum depends on frame of reference but law of conservation of linear momentum is independent of frame of reference.

Newton's laws of motion are valid only in inertial frame of reference.

Weight (w)

It is a field force, the force with which a body is pulled towards the centre of the earth due to gravity. It has the magnitude mg, where m is the mass of the body and g is the acceleration due to gravity.

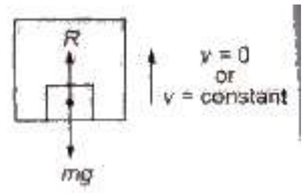
w = mg

Apparent Weight in a Lift

(i) When a lift is at rest or moving with a constant speed, then

R = mg

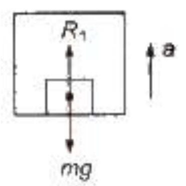
The weighing machine will read the actual weight.



(ii) When a lift is accelerating upward, then apparent weight

$R_1 = m(g + a)$

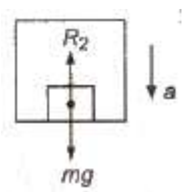
The weighing machine will read the apparent weight, which is more than the actual weight.



(iii) When a lift is accelerating downward, then apparent weight

$R_2 = m(g – a)$

The weighing machine will read the apparent weight, which is less than the actual weight.



(iv) When lift is falling freely under gravity, then

$R_2 = m(g – g)= 0$

The apparent weight of the body becomes zero.

(v) If lift is accelerating downward with an acceleration greater than g, then body will lift from floor to the ceiling of the lift.

Rocket

Rocket is an example of variable mass following law of conservation of momentum.

Thrust on the rocket at any instant $F = – u (dM / dt)$

where u = exhaust speed of the burnt and dM / dt = rate 0f gases combustion of fuel.

Velocity of rocket at any instant is given by $u = v_o + u \log_e (M_o / M)$

where, v_o = initial velocity of the rocket,

M_o = initial mass of the rocket and

M = present mass of the rocket.

If effect of gravity is taken into account then speed of rocket

$u = v_o + u \log_e (M_o / M) – gt$

Friction

A force acting on the point of contact of the objects, which opposes the relative motion is called friction.

It acts parallel to the contact surfaces.

Frictional forces are produced due to intermolecular interactions acting between the molecules of the bodies in contact.

Friction is of three types:

1. **Static Friction**

It is an opposing force which comes into play when one body tends to move over the surface of the other body but actual motion is not taking place.

Static friction is a self adjusting force which increases as the applied force is increased,

2. **Limiting Friction**

It is the maximum value of static friction when body is at the verge of starting motion.

Limiting friction (f_s) = $\mu_s R$

where μ_s, = coefficient of limiting friction and R = normal reaction.

Limiting friction do not depend on area of contact surfaces but depends on their nature, i.e., smoothness or roughness.

If angle of friction is θ, then coefficient of limiting friction

$\mu_s = \tan\theta$

3. **Kinetic Friction**

If the body begins to slide on the surface, the magnitude of the frictional force rapidly decreases to a constant value f_k kinetic friction.

Kinetic friction, $f_k = \mu_k N$

where μ k = coefficient of kinetic friction and N = normal force.

Kinetic friction is of two types:

(a) Sliding friction

(b) Rolling friction

As, rolling friction < sliding friction, therefore it is easier to roll a body than to slide.

Kinetic friction (f_k) = $\mu_k R$

where μ_k = coefficient of kinetic friction and R = normal reaction.

Angle of repose or **angle of sliding** It is the minimum angle of inclination of a plane with the horizontal, such that a body placed on it, just begins to slide down.

If angle of repose is a. and coefficient of limiting friction is μ, then

$\mu_s = \tan\alpha$

Motion on an Inclined Plane

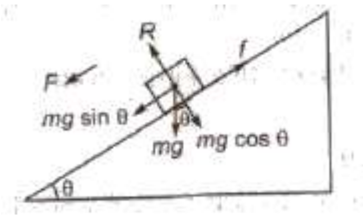
When an object moves along an inclined plane then: different forces act on it like normal reaction of plane, friction force acting in opposite direction of motion etc. Different relations for the motion are given below.

Normal reaction of plane

$R = mg \cos \theta$

and net force acting downward on the block.

$F = mg \sin \theta - f$



Acceleration on inclined plane $a = g (\sin \theta - \mu \cos \theta)$

When angle of inclination of the plane from horizontal is less than the angle of repose (α), then

(i) minimum force required to move the body up the inclined plane

$f_1 = mg (\sin \theta + \mu \cos \theta)$

(ii) minimum force required to push the body down the inclined plane

$f_2 = mg (\mu \cos \theta - \sin \theta)$ J

Tension

Tension force always pulls a body.

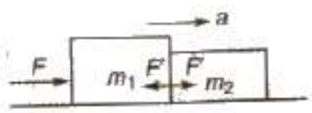
Tension is a reactive force. It is not an active force.

Tension across a massless pulley or frictionless pulley remain constant.

Rope becomes slack when tension force becomes zero.

Motion of Bodies in Contact

(i) **Two Bodies in Contact** If F force is a applied on object of mass m_1 then acceleration of the bodies

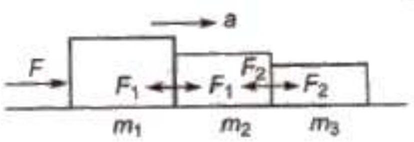


$a = F / (m_1 + m_2)$

Contact force on $m_1 = m_1a = m_1F / (m_1 + m_2)$

Contact force on $m_2 = m_2a = m_2F / (m_1 + m_2)$

(ii) **Three Bodies in Contact** If F force is applied an object of mass m_1, then acceleration of the bodies = $F / (m_1 + m_2 + m_3)$



Contact force between m_1 and m_2

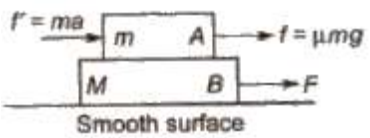
$F_1 = (m_2 + m_3) F / (m_1 + m_2 + m_3)$

Contact force between m_2 and m_3

$F_2 = m_3 F / (m_1 + m_2 + m_3)$

(iii) **Motion of Two Bodies, One Resting on the Other**

(a) The coefficient of friction between surface of A and B be μ. If a force F is applied on the lower body A. then common acceleration of two bodies



$a = F / (M + m)$

Pseudo force acting on block B due to the accelerated motion

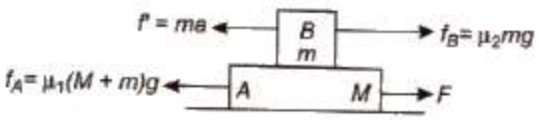
$f' = ma$

The pseudo force tends to produce a relative motion between bodies A and B and consequently a frictional force

$f = \mu N = \mu mg$ is developed. For equilibrium

$ma \le \mu mg$ or $a \le \mu g$

(b) Let friction is also present between the ground surface and body A Let the coefficient of friction between the given surface and body A is μ_1 and the coefficient of friction between the surfaces of bodies A and B is μ_2 If a force F is applied on the lower body A



Net accelerating force = $F - f_A = F - \mu_1(M + m)g$

∴ Net acceleration

$a = F - \mu_1(M + m)g / (M + m) = F / (M + m) - \mu g$

Pseudo force acting on the block B

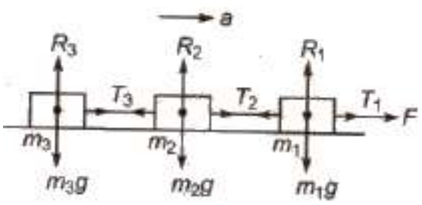
f' = ma

The pseudo force tends to produce a relative motion between the bodies A and B are consequently a frictional force $f_B = \mu$ mg is developed. For equilibrium

ma le; μ_2 mg or $a \leq \mu_2 g$

If acceleration produced under the the effect of force F is more than $\mu_2 g$, then two bodies will not move together.

(iv) **Motion of Bodies Connected by Strings**



Acceleration of the system $a = F / (m_1 + m_2 + m_3)$

Tension in string $T_1 = F$

$T_2 = (m_2 + m_3)\ a = (m_2 + m_3) F / (m_1 + m_2 + m_3)$

$T_3 = m_3 a = m_3 F / (m_1 + m_2 + m_3)$

Pulley Mass System

(i) When unequal masses m_1 and m_2 are suspended from a pulley

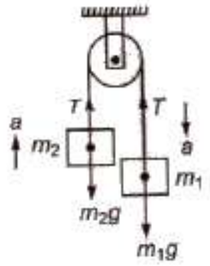
$(m_1 > m_2)$

$m_1 g - T = m_1 a$, and $T - m_2 g = m_2 a$

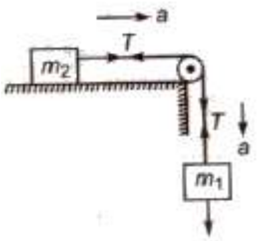
On solving equations, we get

$a = ((m_1 - m_2) / (m_1 + m_2)) * g$

$T = 2m_1m_2 / (m_1 + m_2) * g$



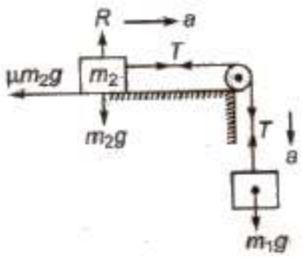
(ii) When a body of mass m_2 is placed frictionless horizontal surface, then



Acceleration $a = m_1g / (m_1 + m_2)$

Tension in string $T = m_1m_2g / (m_1 + m_2)$

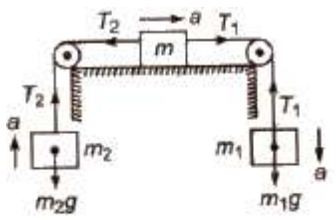
(iii) When a body of mass m2 is placed on a rough horizontal surface, then



Acceleration $a = ((m_1 - \mu m_2) / (m_1 + m_2)) * g$

Tension in string $T = (m_1m_2(1 + \mu) / (m_1 + m_2)) * g$

(iv) When two masses m_1 and m_2 are connected to a single mass M as shown in figure, then



$m_1g - T_1 = m_1a$ …..(i)

$T_2 - m_2g = m_2a$ ……(ii)

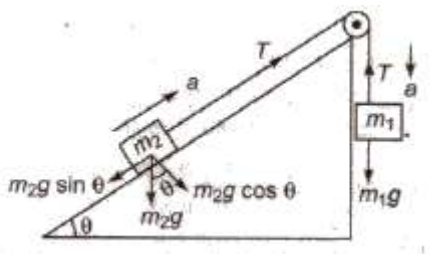
$T_1 - T_2 = Ma$ …….(iii)

Acceleration $a = ((m_1 - m_2 / (m_1 + m_2 + M)) * g$

Tension $T_1 = (2m_2 + M / (m_1 + m_2 + M) * m_1g$

$T_2 = (2m_a + M / (m_1 + m_2 + M) * m_2g$

(v) Motion on a smooth inclined plane, then



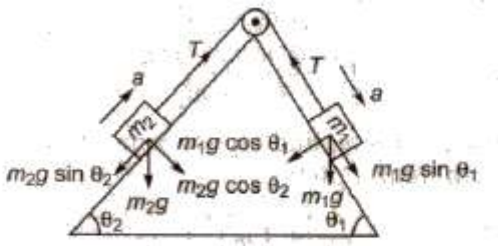
$m_1g - T = m_1a$ …..(i)

$T - m_2g \sin\theta = m_2a$ ……(ii)

Acceleration $a = ((m_1 - m_2 \sin\theta / (m_1 + m_2)) * g$

Tension $T = m_1m_2(1 + \sin\theta)\, g / (m_1 + m_2)$

(vi) Motion of two bodies placed on two inclined planes having different angle of inclination, then



Acceleration $a = (m_1 \sin\theta_1 - m_2 \sin\theta_2)\, g / m_1 + m_2$

Tension $T = (m_1m_2 / m_1 + m_2) * (\sin \theta_1 - \sin \theta_2)\, g$

Physics Notes Class 11 CHAPTER 4 MOTION IN A PLANE part 1

Those physical quantities which require magnitude as well as direction for their complete representation and follows vector laws are called **vectors**.

Vector can be divided into two types

1. Polar Vectors

These are those vectors which have a starting point or a point of application as a displacement, force etc.

2. Axial Vectors

These are those vectors which represent rotational effect and act along the axis of rotation in accordance with right hand screw rule as angular velocity, torque, angular momentum etc.

Scalars

Those physical quantities which require only magnitude but no direction for their complete representation, are called scalars.

Distance, speed, work, mass, density, etc are the examples of scalars. Scalars can be added, subtracted, multiplied or divided by simple algebraic laws.

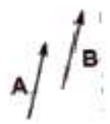
Tensors

Tensors are those physical quantities which have different values in different directions at the same point.

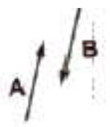
Moment of inertia, radius of gyration, modulus of elasticity, pressure, stress, conductivity, resistivity, refractive index, wave velocity and density, etc are the examples of tensors. Magnitude of tensor is not unique.

Different Types of Vectors

(i) Equal Vectors Two vectors of equal magnitude, in same direction are called equal vectors.



(ii) Negative Vectors Two vectors of equal magnitude but in opposite directions are called negative vectors.



(iii) Zero Vector or Null Vector A vector whose magnitude is zero is known as a zero or null vector. Its direction is not defined. It is denoted by 0.

Velocity of a stationary object, acceleration of an object moving with uniform velocity and resultant of two equal and opposite vectors are the examples of null vector.

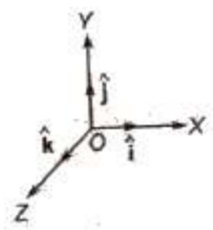
(iv) Unit Vector A vector having unit magnitude is called a unit vector.

A unit vector in the direction of vector **A** is given by

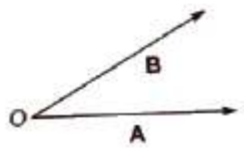
$\hat{A} = A / A$

A unit vector is unitless and dimensionless vector and represents direction only.

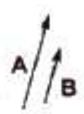
(v) Orthogonal Unit Vectors The unit vectors along the direction of orthogonal axis, i.e., X – axis, Y – axis and Z – axis are called orthogonal unit vectors. They are represented by $\hat{i}, \hat{j}$ and $\hat{k}$



(vi) Co-initial Vectors Vectors having a common initial point, are called co-initial vectors.



(vii) Collinear Vectors Vectors having equal or unequal magnitudes but acting along the same or Ab parallel lines are called collinear vectors.



(viii) Coplanar Vectors Vectors acting in the same plane are called coplanar vectors.

(ix) Localised Vector A vector whose initial point is fixed, is called a localised vector.

(x) Non-localised or Free Vector A vector whose initial point is not fixed is called a non-localised or a free vector.

(xi) Position Vector A vector representing the straight line distance and the direction of any point or object with respect to the origin, is called position vector.

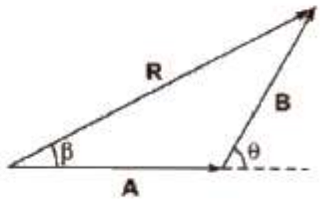
Addition of Vectors

1. Triangle Law of Vectors

If two vectors acting at a point are represented in magnitude and direction by the two sides of a triangle taken in one order, then their resultant is represented by the third side of the triangle taken in the opposite order.

If two vectors A and B acting at a point are inclined at an angle θ, then their resultant

$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$

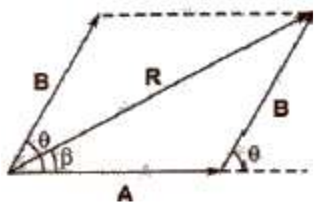


If the resultant vector R subtends an angle β with vector A, then

$\tan \beta = B \sin \theta / A + B \cos \theta$

2. Parallelogram Law of Vectors

If two vectors acting at a point are represented in magnitude and direction by the two adjacent sides of a parallelogram draw from a point, then their resultant is represented in magnitude and direction by the diagonal of the parallelogram draw from the same point.



Resultant of vectors **A** and **B** is given by

$\sqrt{A^2 + B^2 + 2AB \cos \theta}$

If the resultant vector R subtends an angle β with vector A, then

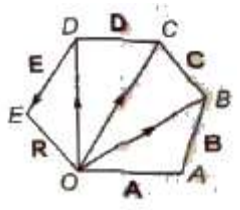
$\tan \beta = B \sin \theta / A + B \cos \theta$

Polygon Law of Vectors

It states that if number of vectors acting on a particle at a time are represented in magnitude and – direction by the various sides of an open polygon taken in same order, their resultant vector E is represented in magnitude and direction by the closing side of polygon taken in opposite order. In fact, polygon law of vectors is the outcome of triangle law of vectors.

R = A + B + C + D + E

OE = OA + AB + BC + CD + DE



Properties of Vector Addition

(i) Vector addition is commutative, i.e., **A + B = B + A**

(ii) Vector addition is associative, i.e.,

A +(B + C)= B + (C + A)= C + (A + B)

(iii) Vector addition is distributive, i.e., m **(A + B) = m A + m B**

Rotation of a Vector

(i) If a vector is rotated through an angle 0, which is not an integral multiple of 2 π, the vector changes.

(ii) If the frame of reference is rotated or translated, the given vector does not change. The components of the vector may, however, change.

Resolution of a Vector into Rectangular Components

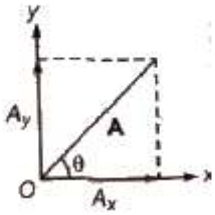
If any vector A subtends an angle θ with x-axis, then its

Horizontal component $A_x = A \cos \theta$

Vertical component $A_y = A \sin \theta$

Magnitude of vector A = $\sqrt{A_x^2 + A_y^2}$

$\tan\theta = A_y / A_x$



Direction Cosines of a Vector

If any vector A subtend angles α, β and γ with x – axis, y – axis and z – axis respectively and its components along these axes are A_x, A_y and A_z, then

$\cos\alpha = A_x / A$, $\cos\beta = A_y / A$, $\cos\gamma = A_z / A$

and $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Subtraction of Vectors

Subtraction of a vector B from a vector A is defined as the addition of vector -B (negative of vector B) to vector A

Thus, **A – B = A + (-B)**

Multiplication of a Vector

1. By a Real Number

When a vector A is multiplied by a real number n, then its magnitude becomes n times but direction and unit remains unchanged.

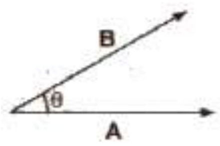
2. By a Scalar

When a vector A is multiplied by a scalar S, then its magnitude becomes S times, and unit is the product of units of A and S but direction remains same as that of vector A.

Scalar or Dot Product of Two Vectors

The scalar product of two vectors is equal to the product of their magnitudes and the cosine of the smaller angle between them. It is denoted by . (dot).

A * B = AB cos θ



The scalar or dot product of two vectors is a scalar.

Properties of Scalar Product

(i) Scalar product is commutative, i.e., **A * B= B * A**

(ii) Scalar product is distributive, i.e., **A * (B + C) = A * B + A * C**

(iii) Scalar product of two perpendicular vectors is zero.

A * B = AB cos 90° = O

(iv) Scalar product of two parallel vectors is equal to the product of their magnitudes, i.e., **A * B** = AB cos 0° = AB

(v) Scalar product of a vector with itself is equal to the square of its magnitude, i.e.,

A * A = AA cos 0° = A^2

(vi) Scalar product of orthogonal unit vectors

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$$

and $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{i}} = 0$

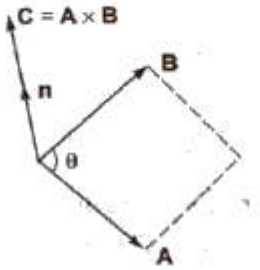
(vii) Scalar product in cartesian coordinates

$$\mathbf{A} \cdot \mathbf{B} = (A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}) \cdot (B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}})$$

$= A_xB_x + A_yB_y + A_zB_z$

Vector or Cross Product of Two Vectors

The vector product of two vectors is equal to the product of their magnitudes and the sine of the smaller angle between them. It is denoted by * (cross).



A * B = AB sin θ n

The direction of unit vector n can be obtained from right hand thumb rule.

If fingers of right hand are curled from A to B through smaller angle between them, then thumb will represent the direction of vector (A * B).

The vector or cross product of two vectors is also a vector.

Properties of Vector Product

(i) Vector product is not commutative, i.e.,

A * B ≠ B * A [∴ (A * B) = — (B * A)]

(ii) Vector product is distributive, i.e.,

A * (B + C) = A * B + A * C

(iii) Vector product of two parallel vectors is zero, i.e.,

A * B = AB sin O° = 0

(iv) Vector product of any vector with itself is zero.

A * A = AA sin O° = 0

(v) Vector product of orthogonal unit vectors

$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$

and $\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}$

(vi) Vector product in cartesian coordinates

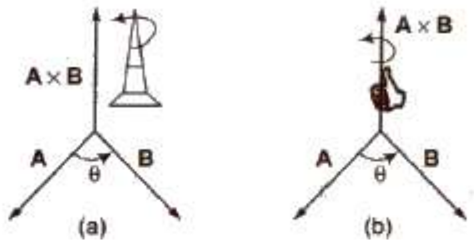
$$\mathbf{A} \times \mathbf{B} = (A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}) \times (B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}})$$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_yB_z - A_zB_y)\,\hat{\mathbf{i}} - (A_xB_z - B_xA_z)\,\hat{\mathbf{j}} + (A_xB_y - A_yB_x)\,\hat{\mathbf{k}}$$

Direction of Vector Cross Product

When C = A * B, the direction of C is at right angles to the plane containing the vectors A and B. The direction is determined by the right hand screw rule and right hand thumb rule.



(i) Right Hand Screw Rule Rotate a right handed screw from first vector (A) towards second vector (B). The direction in which the right handed screw moves gives the direction of vector (C).

(ii) Right Hand Thumb Rule Curl the fingers of your right hand from A to B. Then, the direction of the erect thumb will point in the direction of A * B.

Physics Notes Class 11 CHAPTER 3 MOTION IN A STRAIGHT LINE

Motion

If an object changes its position with respect to its surroundings with time, then it is called in motion.

Rest

If an object does not change its position with respect to its surroundings with time, then it is called at rest.

[Rest and motion are relative states. It means an object which is at rest in one frame of reference can be in motion in another frame of reference at the same time.]

Point Mass Object An object can be considered as a point mass object, if the distance travelled by it in motion is very large in comparison to its dimensions.

Types of Motion

1. One Dimensional Motion

If only one out of three coordinates specifying the position of the object changes with respect to time, then the motion is called one dimensional motion.

For instance, motion of a block in a straight line motion of a train along a straight track a man walking on a level and narrow road and object falling under gravity etc.

2. Two Dimensional Motion

If only two out of three coordinates specifying the position of the object changes with respect to time, then the motion is called two dimensional motion.

A circular motion is an instance of two dimensional motion.

3. Three Dimensional Motion

If all the three coordinates specifying the position of the object changes with respect to time, then the motion is called three dimensional motion.

A few instances of three dimension are flying bird, a flying kite, a flying aeroplane, the random motion of gas molecule etc.

Distance

The length of the actual path traversed by an object is called the distance.

It is a scalar quantity and it can never be zero or negative during the motion of an object.

Its unit is metre.

Displacement

The shortest distance between the initial and final positions of any object during motion is called displacement. The displacement of an object in a given time can be positive, zero or negative.

It is a vector quantity.

Its unit is metre.

Speed

The time rate of change of position of the object in any direction is called speed of the object.

Speed (v) = Distance travelled (s) / Time taken (t)

Its unit is m/s.

It is a scalar quantity.

Its dimensional formula is $[M^oT^{-1}]$.

Uniform Speed

If an object covers equal distances in equal intervals of time, then its speed is called uniform speed.

Non-uniform or Variable Speed

If an object covers unequal distances in equal intervals of time, then its speed is called non-uniform or variable speed.

Average Speed

The ratio of the total distance travelled by the object to the total time taken is called average speed of the object.

Average speed = Total distanced travelled / Total time taken

If a particle travels distances s_1, s_2, s_3 , … with speeds v_1, v_2, v_3, …, then

Average speed = $s_1 + s_2 + s_3 + \ldots / (s_1 / v_1 + s_2 / v_2 + s_3 / v_3 + \ldots)$

If particle travels equal distances ($s_1 = s_2 = s$) with velocities v_1 and v_2, then

Average speed = $2 v_1 v_2 / (v_1 + v_2)$

If a particle travels with speeds v_1, v_2, v_3, …, during time intervals t_1, t_2, t_3,…, then

Average speed = $v_1t_1 + v_2t_2 + v_3t_3 + \ldots / t_1 + t_2 + t_3 + \ldots$

If particle travels with speeds v_1, and v_2 for equal time intervals, i.e., $t_1 = t_2 = t_3$, then

Average speed = $v_1 + v_2 / 2$

When a body travels equal distance with speeds V_1 and V_2, the average speed (v) is the harmonic mean of two speeds.

$2 / v = 1 / v_1 + 1 / v_2$

Instantaneous Speed

When an object is travelling with variable speed, then its speed at a given instant of time is called its instantaneous speed.

Instantaneous speed = $\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$

Velocity

Tlie rate of change of displacement of an object in a particular direction is called its velocity.

Velocity = Displacement / Time taken

Its unit is m/s.

Its dimensional formula is $[M^oT^{-1}]$.

It is a vector quantity, as it has both, the magnitude and direction.

The velocity of an object can be positive, zero and negative.

Uniform Velocity

If an object undergoes equal displacements in equal intervals of time, then it is said to be moving with a uniform velocity.

Non-uniform or Variable Velocity

If an object undergoes unequal displacements in equal intervals of time, then it is said to be moving with a non-uniform or variable velocity.

Relative Velocity

Relative velocity of one object with respect to another object is the time rate of change of relative position of one object with respect to another object.

Relative velocity of object A with respect to object B

$V_{AB} = V_A - V_B$

When two objects are moving in the same direction, then

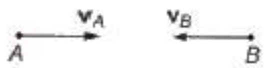
$\mathbf{v}_{AB} = \mathbf{v}_A - \mathbf{v}_B$

or $v_{AB} = v_A - v_B$

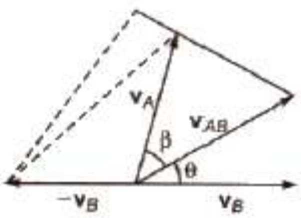
When two objects are moving in opposite direction, then

$\mathbf{v}_{AB} = \mathbf{v}_A + \mathbf{v}_B$

or $v_{AB} = v_A + v_B$



When two objects are moving at an angle, then



$v_{AB} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos\theta}$

and $\tan\beta = v_B \sin\theta / v_A - v_B \cos\theta$

Average Velocity

The ratio of the total displacement to the total time taken is called average velocity.

Average velocity = Total displacement / Total time taken

Acceleration

The time rate of change of velocity is called acceleration.

Acceleration (a) = Change in velocity (Δv) / Time interval (Δt)

Its unit is m/s^2

Its dimensional formula is $[M^oLT^{-2}]$.

It is a vector quantity.

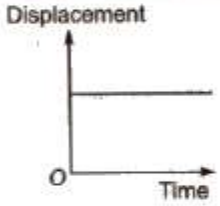
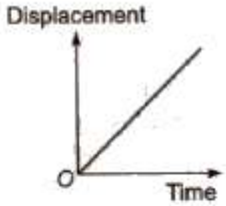
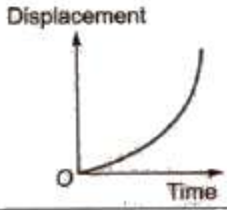
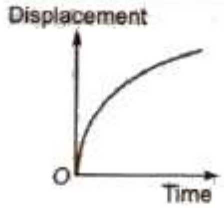
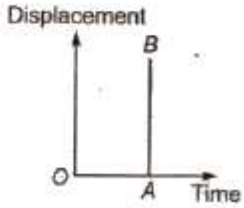
Acceleration can be positive, zero or negative. Positive acceleration means velocity increasing with time, zero acceleration means velocity is uniform while negative acceleration (retardation) means velocity is decreasing with time.

If a particle is accelerated for a time t_1 with acceleration a_1 and for a time t_2 with acceleration a_2, then average acceleration

$a_{av} = a_1t_1 + a_2t_2 / t_1 + t_2$

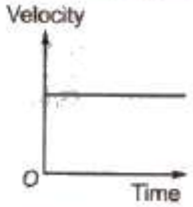
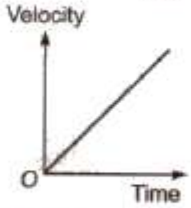
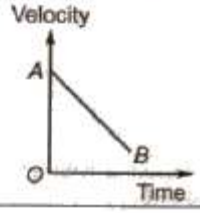
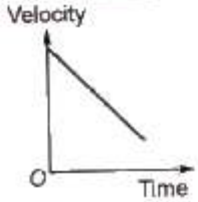
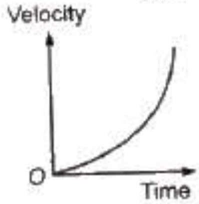
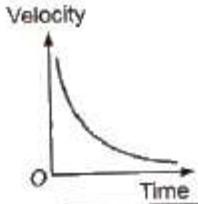
Different Graphs of Motion

Displacement – Time Graph

S. No.	Condition	Graph
(a)	For a stationary body	
(b)	Body moving with a constant velocity	
(c)	Body moving with a constant acceleration	
(d)	Body moving with a constant retardation	
(e)	Body moving with infinite velocity. But such motion of a body is never possible.	

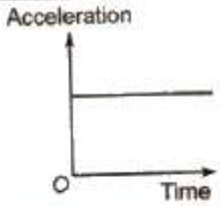
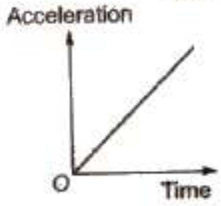
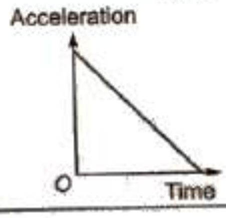
Note Slope of displacement-time graph gives average velocity.

Velocity – Time Graph

S. No.	Condition	Graph
(a)	Moving with a constant velocity	
(b)	Moving with a constant acceleration	
(c)	Body moving with a constant retardation and its initial velocity is not zero.	
(d)	Moving with a constant retardation	
(e)	Moving with increasing acceleration	
(f)	Moving with decreasing acceleration	

Note Slope of velocity-time graph gives average acceleration.

Acceleration – Time

S. No.	Condition	Graph
(a)	When object is moving with constant acceleration	
(b)	When object is moving with constant increasing acceleration	
(c)	When object is moving with constant decreasing acceleration	

Equations of Uniformly Accelerated Motion

If a body starts with velocity (u) and after time t its velocity changes to v, if the uniform acceleration is a and the distance travelled in time t in s, then the following relations are obtained, which are called equations of uniformly accelerated motion.

(i) $v = u + at$

(ii) $s = ut + at^2$

(iii) $v^2 = u^2 + 2as$

(iv) Distance travelled in nth second.

$S_n = u + a / 2(2n – 1)$

If a body moves with uniform acceleration and velocity changes from u to v in a time interval, then the velocity at the mid point of its path

$\sqrt{}u^2 + v^2 / 2$

Motion Under Gravity

If an object is falling freely (u = 0) under gravity, then equations of motion

(i) $v = u + gt$

(ii) $h = ut + gt^2$

(iii) $V^2 = u^2 + 2gh$

Note If an object is thrown upward then g is replaced by – g in above three equations.

It thus follows that

(i) Time taken to reach maximum height

$t_A = u / g = \sqrt{2h / g}$

(ii) Maximum height reached by the body

$h_{max} = u^2 / 2g$

(iii) A ball is dropped from a building of height h and it reaches after t seconds on earth. From the same building if two ball are thrown (one upwards and other downwards) with the same velocity u and they reach the earth surface after t, and t2 seconds respectively, then

$t = \sqrt{t_1 t_2}$

(iv) When a body is dropped freely from the top of the tower and another body is projected horizontally from the same point, both will reach the ground at the same time.

Physics Notes Class 11 CHAPTER 2 UNITS AND MEASUREMENTS

The comparison of any physical quantity with its standard unit is called **measurement.**

Physical Quantities

All the quantities in terms of which laws of physics are described, and whose measurement is necessary are called physical quantities.

Units

- A definite amount of a physical quantity is taken as its standard unit.
- The standard unit should be easily reproducible, internationally accepted.

Fundamental Units

Those physical quantities which are independent to each other are called fundamental quantities and their units are called fundamental units.

S.No.	Fundamental Quantities	Fundamental Units	Symbol
1.	Length	metre	m
2.	Mass	kilogram	kg
3.	Time	second	S
4.	Temperature	kelvin	kg
5	Electric current	ampere	A
6	Luminous intensity	candela	cd
7	Amount of substance	mole	mol

Supplementary Fundamental Units

Radian and steradian are two supplementary fundamental units. It measures plane angle and solid angle respectively.

S.No.	Supplementary Fundamental Quantities	Supplementary Unit	Symbol
1	Plane angle	radian	rad
2	Solid angle	steradian	Sr

Derived Units

Those physical quantities which are derived from fundamental quantities are called derived quantities and their units are called derived units.
e.g., velocity, acceleration, force, work etc.

Definitions of Fundamental Units

The seven fundamental units of SI have been defined as under.

1. **1 kilogram** A cylindrical prototype mass made of platinum and iridium alloys of height 39 mm and diameter 39 mm. It is mass of 5.0188 x 10^{25} atoms of carbon-12.
2. **1 metre** 1 metre is the distance that contains 1650763.73 wavelength of orange-red light of Kr-86.
3. **1 second** 1 second is the time in which cesium atom vibrates 9192631770 times in an atomic clock.
4. **1 kelvin** 1 kelvin is the (1/273.16) part of the thermodynamics temperature of the triple point of water.
5. **1 candela** 1 candela is (1/60) luminous intensity of an ideal source by an area of cm' when source is at melting point of platinum (1760°C).
6. **1 ampere** 1 ampere is the electric current which it maintained in two straight parallel conductor of infinite length and of negligible cross-section area placed one metre apart in vacuum will produce between them a force 2 x 10^{-7} N per metre length.
7. **1 mole** 1 mole is the amount of substance of a system which contains a many elementary entities (may be atoms, molecules, ions, electrons or group of particles, as this and atoms in 0.012 kg of carbon isotope $_6C^{12}$.

Systems of Units

A system of units is the complete set of units, both fundamental and derived, for all kinds of physical quantities. The common system of units which is used in mechanics are given below:

1. **CGS System** In this system, the unit of length is centimetre, the unit of mass is gram and the unit of time is second.
2. **FPS System** In this system, the unit of length is foot, the unit of mass is pound and the unit of time is second.
3. **MKS System** In this system, the unit of length is metre, the unit of mass is kilogram and the unit of time is second.
4. **SI System** This system contain seven fundamental units and two supplementary fundamental units.

Relationship between Some Mechanical SI Unit and Commonly Used Units

S.No.	Physical Quantity	Unit
1	Length	(a) 1 micrometre = 10^{-6} m (b) 1 angstrom =10^{-10} m
2	Mass	(a) 1 metric ton = 10^3 kg

		(b) 1 pound = 0.4537 kg
		(c) 1 amu = 1.66 x10^{-23} kg
3	Volume	1 litre = 10^{-32} m^3
4.	Force	(a) 1 dyne = 10^{-5} N (b) 1 kgf = 9.81 N
5.	Pressure	(a) 1 kgfm^2 = 9.81Nm^{-2} (b) 1 mm of Hg = 133 Nm^{-2} (c) 1 pascal = 1 Nm^{-2} (d) 1 atmosphere pressure = 76 cm of Hg = 1.01 x 10^5 pascal
6.	Work and energy	(a) 1 erg =10^{-7} J (b) 1 kgf-m = 9.81 J (c) 1 kWh = 3.6 x 10^6 J (d) 1 eV = 1.6 x 10^{-19} J
7.	Power	(d) 1 kgf- ms^{-1} = 9.81W 1 horse power = 746 W

Some Practical Units

1. 1 fermi =10^{-15} m
2. 1 X-ray unit = 10^{-13} m
3. 1 astronomical unit = 1.49 x 10^{11} m (average distance between sun and earth)
4. 1 light year = 9.46 x 10^{15} m
5. 1 parsec = 3.08 x 10^{16} m = 3.26 light year

Some Approximate Masses

Object	**Kilogram**
Our galaxy	2 x 10^{41}
Sun	2 x 10^{30}
Moon	7 x 10^{22}
Asteroid Eros	5 x 10^{15}

Dimensions

Dimensions of any physical quantity are those powers which are raised on fundamental units to express its unit. The expression which shows how and which of the base quantities represent the dimensions of a physical quantity, is called the dimensional formula.

Dimensional Formula of Some Physical Quantities

S.No.	**Physical** Quantity	**Dimensional** Formula	**MKS** Unit

1	Area	$[L^2]$	metre^2
2	Volume	$[L^3]$	metre^3
3	Velocity	$[LT^{-1}]$	ms^{-1}
4	Acceleration	$[LT^{-2}]$	ms^{-2}
5	Force	$[MLT^{-2}]$	newton (N)
6	Work or energy	$[ML^2T^{-2}]$	joule (J)
7	Power	$[ML^2T^{-3}]$	J s^{-1} or watt
8	Pressure or stress	$[ML^{-1}T^{-2}]$	Nm^{-2}
9	Linear momentum or Impulse	$[MLT^{-1}]$	kg ms^{-1}
10	Density	$[ML^{-3}]$	kg m^{-3}
11	Strain	Dimensionless	Unitless
12	Modulus of elasticity	$[ML^{-1}T^{-2}]$	Nm^{-2}
13	Surface tension	$[MT^{-2}]$	Nm^{-1}
14	Velocity gradient	T^{-1}	second^{-1}
15	Coefficient of velocity	$[ML^{-1}T^{-1}]$	kg $m^{-1}s^{-1}$
16	Gravitational constant	$[M^{-1}L^3T^{-2}]$	Nm^2/kg^2
17	Moment of inertia	$[ML^2]$	kg m^2
18	Angular velocity	$[T^{-1}]$	rad/s
19	Angular acceleration	$[T^{-2}]$	rad/S^2
20	Angular momentum	$[ML^2T^{-1}]$	kg m^2S^{-1}
21	Specific heat	$L^2T^{-2}\theta^{-1}$	kcal $kg^{-1}K^{-1}$
22	Latent heat	$[L^2T^{-2}]$	kcal/kg
23	Planck's constant	ML^2T^{-1}	J^{-s}
24	Universal gas constant	$[ML^2T^{-2}\theta^{-1}]$	J/mol-K

Homogeneity Principle

If the dimensions of left hand side of an equation are equal to the dimensions of right hand side of the equation, then the equation is dimensionally correct. This is known as **homogeneity principle.**

Mathematically [LHS] = [RHS]

Applications of Dimensions

1. To check the accuracy of physical equations.
2. To change a physical quantity from one system of units to another system of units.
3. To obtain a relation between different physical quantities.

Significant Figures

In the measured value of a physical quantity, the number of digits about the correctness of which we are sure plus the next doubtful digit, are called the significant figures.

Rules for Finding Significant Figures

1. All non-zeros digits are significant figures, e.g., 4362 m has 4 significant figures.
2. All zeros occuring between non-zero digits are significant figures, e.g., 1005 has 4 significant figures.
3. All zeros to the right of the last non-zero digit are not significant, e.g., 6250 has only 3 significant figures.
4. In a digit less than one, all zeros to the right of the decimal point and to the left of a non-zero digit are not significant, e.g., 0.00325 has only 3 significant figures.
5. All zeros to the right of a non-zero digit in the decimal part are significant, e.g., 1.4750 has 5 significant figures.

Significant Figures in Algebric Operations

(i) In Addition or Subtraction In addition or subtraction of the numerical values the final result should retain the least decimal place as in the various numerical values. e.g.,

If l_1= 4.326 m and l_2 = 1.50 m

Then, $l_1 + l_2$ = (4.326 + 1.50) m = 5.826 m

As l_2 has measured upto two decimal places, therefore

$l_1 + l_2$ = 5.83 m

(ii) In Multiplication or Division In multiplication or division of the numerical values, the final result should retain the least significant figures as the various numerical values. e.g., If length 1= 12.5 m and breadth b = 4.125 m.

Then, area A = l x b = 12.5 x 4.125 = 51.5625 m^2

As l has only 3 significant figures, therefore

A= 51.6 m^2

Rules of Rounding Off Significant Figures

1. If the digit to be dropped is less than 5, then the preceding digit is left unchanged. e.g., 1.54 is rounded off to 1.5.
2. If the digit to be dropped is greater than 5, then the preceding digit is raised by one. e.g., 2.49 is rounded off to 2.5.
3. If the digit to be dropped is 5 followed by digit other than zero, then the preceding digit is raised by one. e.g., 3.55 is rounded off to 3.6.
4. If the digit to be dropped is 5 or 5 followed by zeros, then the preceding digit is raised by one, if it is odd and left unchanged if it is even. e.g., 3.750 is rounded off to 3.8 and 4.650 is rounded off to 4.6.

Error

The lack in accuracy in the measurement due to the limit of accuracy of the instrument or due to any other cause is called an error.

1. Absolute Error

The difference between the true value and the measured value of a quantity is called absolute error.

If a_1 , a_2, a_3 ,…, a_n are the measured values of any quantity a in an experiment performed n times, then the arithmetic mean of these values is called the true value (a_m) of the quantity.

$$a_m = \frac{a_1 + a_2 + a_3 + \ldots + a_n}{n}$$

The absolute error in measured values is given by

$\Delta a_1 = a_m - a_1$
$\Delta a_2 = a_m - a_1$

…………

$\Delta a_m = \Delta a_m - \Delta a_n$

2. Mean Absolute Error

The arithmetic mean of the magnitude of absolute errors in all the measurement is called mean absolute error.

$$\overline{\Delta a} = \frac{|\Delta a_1| + |\Delta a_2| + \ldots + |\Delta a_n|}{n}$$

3. Relative Error The ratio of mean absolute error to the true value is called relative

$$\text{Relative error} = \frac{\text{Mean absolute error}}{\text{True value}} = \frac{\overline{\Delta a}}{a_m}$$

4. Percentage Error The relative error expressed in percentage is called percentage error.

$$\text{Percentage error} = \frac{\overline{\Delta a}}{a_m} \times 100\%$$

Propagation of Error

(i) Error in Addition or Subtraction Let x = a + b or x = a – b

If the measured values of two quantities a and b are (a ± Δa and (b ± Δb), then maximum absolute error in their addition or subtraction.

$\Delta x = \pm(\Delta a + \Delta b)$

(ii) Error in Multiplication or Division Let x = a x b or x = (a/b).
If the measured values of a and b are (a ± Δa) and (b ± Δb), then maximum relative error

$$\frac{\Delta x}{x} = \pm\left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right)$$

Chapter 1
Physical World

Science and its origin

Science is a **systematic understanding of natural phenomena** in detail so that it can be **predicted, controlled and modified**. Science involves exploring, experimenting and speculating phenomena happening around us.

- The word **Science** is derived from a latin verb *Scientia* meaning 'to know'.
- **Scientific method** is a way to gain knowledge in a systematic and in-depth way. It involves:
  - *Systematic observations*
  - *Controlled experiments*
  - *Qualitative and Quantitative reasoning*
  - *Mathematical modeling*
  - *Prediction and verification (or falsification) of theories*
  - *Speculation or Prediction*
- Science does not have any final theory. The improved observations, accurate tools keep improving the knowledge and perspective. Johannes Kepler used Tycho Brahe's research on planetary motion to improve Nicolas Copernicus theory.
- Quantum mechanics was developed to deal with atomic and nuclear phenomena. Work of Ernest Rutherford on nuclear model of atom became basis of quantum theory given by Niels Bohr. Antiparticle theory of Paul Dirac led to the discovery of antielectron (positron) by Carl Anderson.

Natural Sciences

Natural science is a branch of science concerned with the description, prediction, and understanding of natural phenomena, based on observational and empirical evidence. It consists of following disciplines:

- Physics
- Chemistry
- Biology

Physics

Physics is a study of basic laws of nature and their manifestation in different natural phenomena. Physics is the study of physical world and matter and its motion through space and time, along with related concepts such as energy and force.

- Word Physics is derived from a Greek word *phusikḗ* meaning nature.
- Two principal types of approaches in Physics are:
  1. **Unification**: This approach considers all of the world's phenomena as a collection of universal laws in different domains and conditions. Example, law of gravitation applies both to a falling apple from a tree as well as motion of planets around the sun. Electromagnetism laws govern all electric and magnetic phenomena.
  2. **Reduction**: This approach is to derive properties of complex systems from the properties and interaction of its constituent parts. Example, temperature studied under thermodynamics is also related to average kinetic energy of molecules in a system (kinetic theory).

Impact and uses of Physics:

- It can explain a phenomena happening over a large magnitude with a simple theory.
- Experiments and observations are used to develop new theories for unidentified phenomena and improve old theories for existing phenomena.
- Development of devices using laws of physics.

Scope of Physics

Scope of Physics is vast as it covers quantities with length magnitude as high as 10^{40}m or more (astronomical studies of universe) and as low as 10^{-14}m or less (study of electrons, protons etc). Similarly the range of time scale goes from 10^{-22}s to 10^{18}s and mass from 10^{-30}kg to 10^{55}kg.

Physics is broadly divided into two types based on its scope - **Classical Physics and Modern Physics**. Classical physics deal with the **macroscopic phenomena** while the modern physics deals with the **microscopic phenomena**.

Macroscopic Domain

Macroscopic domain includes phenomena at large scales like laboratory, terrestrial and astronomical. It includes following subjects:

1. **Mechanics** – It is based on Newton's laws on motion and the laws of gravitation. It is concerned with motion/equilibrium of particles, rigid and deformable bodies and general system of particles. Examples,
    a. Propulsion of rocket by ejecting gases
    b. Water/Sound waves
    c. Equilibrium of bent rod under a load
2. **Electrodynamics** – It deals with electric and magnetic phenomena associated with charged and magnetic bodies. Examples,
    a. motion of a current-carrying conductor in a magnetic field
    b. the response of a circuit to an ac voltage (signal)
    c. the propagation of radio waves in the ionosphere
3. **Optics** – It deals with phenomena involving light. Examples,
    a. Reflection and refraction of light
    b. Dispersion of light through a prism
    c. Colour exhibited by thin films
4. **Thermodynamics** – It deals with systems in macroscopic equilibrium and changes in internal energy, temperature, entropy etc. of systems under application of external force or heat. Examples,
    a. Efficiency of heat engines
    b. Direction of physical and chemical process

Microscopic Domain

Microscopic domain includes phenomena at minuscule scales like atomic, molecular and nuclear. It also deals with interaction of probes like electrons, photons and other elementary particles. Quantum theory has been developed to handle these phenomena.

Factors responsible for progress of Physics

- Quantitative analysis along with qualitative analysis.
- Application of universal laws in different contexts.
- Approximation approach (complex phenomena broken down into collection of basic laws).
- Extracting and focusing on essential features of a phenomenon.

Hypothesis, Axiom and Models

a) **Hypothesis** is a supposition without assuming that it is true. It may not be proved but can be verified through a series of experiments.
b) **Axiom** is a self-evident truth that it is accepted without controversy or question.
c) **Model** is a theory proposed to explain observed phenomena.
d) **Assumption** is the basis of physics, where a number of phenomena can be explained. These assumptions are made from experiments, observation and a lot of statistical data.

Technological applications of Physics

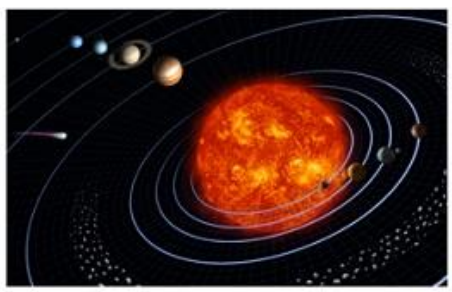
Several examples where Physics and its concepts have led to discoveries/inventions are listed below.

- Steam engine was developed from the industrial revolution in eighteenth century.
- Wireless communication was developed after discovery of laws of electricity and magnetism.
- Neuron-induced fission of uranium, done by Hahn and Meitner in 1938, led to the formation of nuclear power reactors and nuclear weapons.
- Conversion of solar, wind, geothermal etc. energy into electricity.

Fundamental Forces in nature

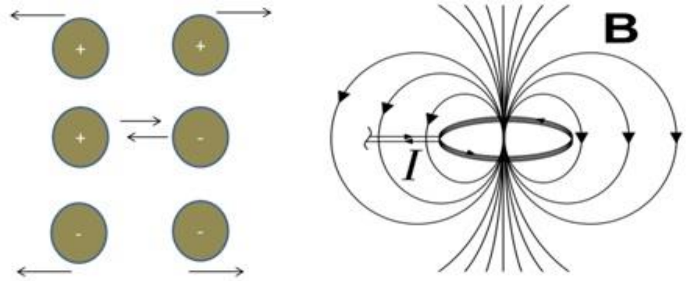
The forces which we see in our day to day life like muscular, friction, forces due to compression and elongation of springs and strings, fluid and gas pressure, electric, magnetic, interatomic and intermolecular forces are **derived forces** as their originations are due to a few fundamental forces in nature.
A few fundamental forces are:

1. **Gravitational Force**: It is the **force of mutual attraction** between any two objects by virtue of their masses. It is a **universal force** as every object experiences this force due to every other object in the universe.



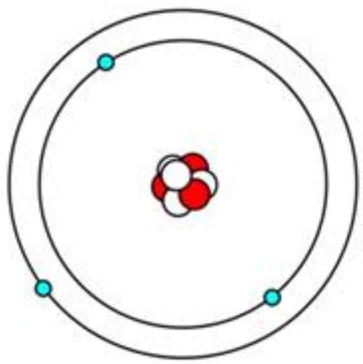
The gravitational force causes the apple to fall as well as planets to revolve around the sun.

2. **Electromagnetic Force**: It is the **force between charged particles**. Charges at rest have electric attraction (between unlike charges) and repulsion (between like charges). Charges in motion produce magnetic force. Together they are called Electromagnetic Force.



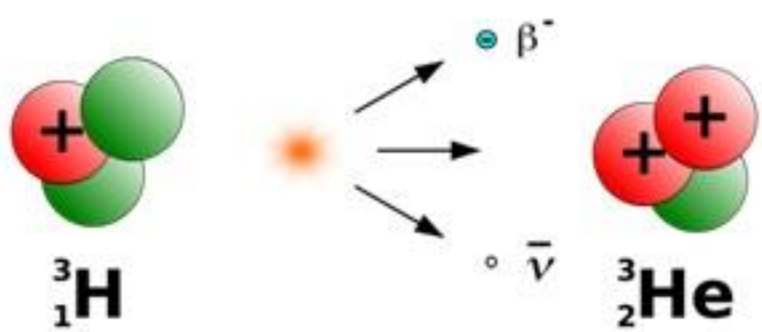
The unlike charges attract each other while like charges repel each other. A current carrying wire generates magnetic field around it, giving rise to electromagnetism.

3. **Strong Nuclear Force**: It is the **attractive force between protons and neutrons** in a nucleus.It is charge-independent and acts equally between a proton and a proton, a neutron and a neutron, and a proton and a neutron. Recent discoveries show that protons and neutrons are built of elementary particles, **quarks**.



Bohr Model of Lithium Atom. Nucleus is a tightly bound entity with a strong force of attraction between protons and neutrons. Electrons revolve around the nucleus.

4. **Weak Nuclear Force:** This force appears only in certain nuclear processes such as the **β-decay of a nucleus**. In β-decay, the nucleus emits an electron and an uncharged particle called **neutrino**.This particle was first predicted by Wolfgang Pauli in 1931.



B-decay of a nucleus.

Below table shows difference between the above forces.

Name	Relative Strength	Range	Operates among
Gravitational force	10^{-39}	Infinite	All objects in the universe

Weak nuclear force	10^{-13}	Very short, Sub-nuclearsize (-10^{-16}m)	Some elementary particles, particularly electron and neutrino
Electromagnetic force	10^{-2}	Infinite	Charged particles
Strong nuclear force	1	Short, nuclear size (-10^{-15}m)	Nucleons, heavier elementary particles

5. **<u>Unification of Forces</u>:** There have been physicists who have tried to combine a few of the above fundamental forces. These are listed in table below.

Name of Physicist	**Year**	**Achievement in Unification**
Isaac Newton	1687	Unified celestial and terrestrial mechanics.
Hans Christian Oersted and Michael Faraday	1820 and 1830 respectively	Unified electric and magnetic phenomena to give rise to electromagnetism.
James Clerk Maxwell	1873	Unified electricity, magnetism and optics to show that light is an electromagnetic wave.
Sheldon Glashow, Abdus Salam, Steven Weinberg	1979	Gave the idea of electro-weak force which is a combination of electromagnetic and weak nuclear force.
Carlo Rubia, Simon Vander Meer	1984	Verified the theory of elctro-weak force.

<u>Conserved Quantities</u>

Physics gives laws to summarize the investigations and observations of the phenomena occurring in the universe.

- Physical quantities that remain constant with time are called **conserved quantities**. Example, for a body under external force, the kinetic and potential energy change over time but the total mechanical energy (kinetic + potential) remains constant.
- Conserved quantities can be scalar (Energy) or vector (Total linear momentum and total angular momentum).

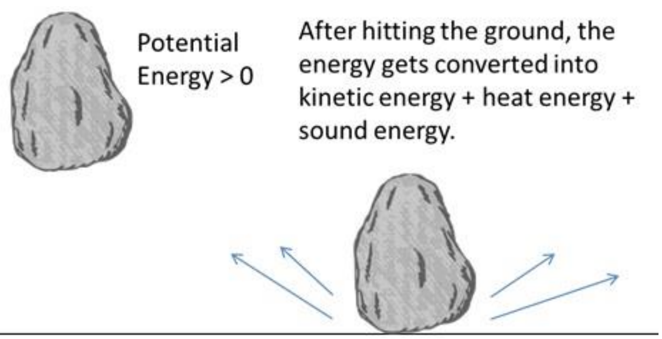
A ball in air falling to ground has some Potential energy and zero Kinetic energy. As soon as it touches the ground the potential energy gets converted into kinetic energy. The total Mechanical energy remains same.

<u>Conservation Laws</u>

A **conservation law** is a hypothesis based on observation and experiments which cannot be proved. These can be verified via experiments.

<u>Law of conservation of Energy</u>

- According to the **general Law of conservation of energy**, the energies remain constant over time and convert from one form to another.
- The law of conservation of energy applies to the whole universe and it is believed that the total energy of the universe remains unchanged.
- Under identical conditions, the nature produces symmetric results at different time.



<u>Law of conservation of Mass</u>

This is a principle used in analysis of chemical reactions.

- A **chemical reaction** is basically a rearrangement of atoms among different molecules.
- If the total binding energy of the reacting molecules is less than the total binding energy of the product molecules, the difference appears as heat and the reaction is **exothermic**.
- The opposite is true for energy absorbing (**endothermic**) reactions.
- Since the atoms are merely rearranged but not destroyed, the **total mass of the reactants is the same as the total mass of the products in a chemical reaction**.
- Mass is related to energy through Einstein theory,

$\mathbf{E = mc^2}$, where c is the speed of light in vacuum

<u>Law of conservation of linear momentum</u>

- Symmetry of laws of nature with respect to translation in space is termed as law of conservation of linear momentum.
- Example law of gravitation is same on earth and moon even if the acceleration due to gravity at moon is $1/6^{th}$ than that at earth.

<u>Law of conservation of angular momentum</u>

- Isotropy of space (no intrinsically preferred direction in space) underlies the law of conservation of angular momentum.